Fiber steering decouples nonlinear hardening from linear frequency in composite panels

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Abstract

Question. Thin composite panels harden as they vibrate: the resonance drifts upward with amplitude and, under forcing, folds and jumps. Can the *spatial distribution* of fiber orientation reduce this hardening at fixed linear frequency and mass, by what mechanism, and how does it compare with a uniform ply angle and with the stacking sequence?
Hypothesis. The frequency and the hardening integrate the same rotated stiffness against two different strain fields of the fundamental mode, its curvature and its in-plane-relaxed stretch. Where these fields differ in principal direction, a fiber field can serve both pointwise; a uniform layup can only compromise.
Analysis. A local Fourier representation predicts no lever for isotropic material, invariance of a single-ply beam under any uniform angle, and a pointwise first-order optimality condition. We test the predictions on laminated von Kármán beams and plates, with the frequency-normalized hardening J of a spectral-submanifold (SSM) reduction (in-plane relaxation exact for the von Kármán model, mode-adaptive, analytic adjoint), the frequency as an equality constraint and the tow-curvature limit imposed pointwise.
Results. On a clamped beam, symmetric steering cuts the hardening by up to 59% by placing compliant fibers at the inflection points. On a simply supported plate, a curvature-feasible idealized tow field ($R_{\min} = 20\,\mathrm{mm}$) brings J to 0.366 of the unidirectional value at the unidirectional frequency; with the frequency re-matched in refined and independent models the ratio is 0.352–0.370. At that frequency no other uniform symmetric laminate exists, and above it steering undercuts every uniform class surveyed. The lever vanishes for isotropic material, grows with anisotropy, saturates once tows turn on the strain-field length scale, and persists for clamped edges (0.40 against 0.89 for the best uniform laminate), a square plate (0.22) and in-plane movable edges (0.49); it is weak on a 2:1 plate whose near-cylindrical mode leaves little conflict between the two fields. The jump-free amplitude widens by 1.66×, as $a_c \propto J^{-1/2}$ predicts.
Conclusion. Spatial fiber steering is a mechanistically understood lever on nonlinear stiffness at fixed linear frequency and mass, set by the material anisotropy, the conflict between the two strain fields and the tow radius. Laying the idealized fields requires tow cut/add; an as-built defect model keeps J/J_{UD} between 0.29 and 0.38, but no longer at the held frequency and mass (frequency 0.96 to 1.12, mass 0.97 to 1.12 of UD).

Keywords: variable-stiffness composites; fiber steering; geometric nonlinearity; spectral submanifolds; backbone curve; adjoint optimization; automated fiber placement.

1 Introduction

Motivation. Thin composite panels stiffen as they vibrate. Mid-plane stretching makes the restoring force of a bending mode cubic in its amplitude, so the resonance frequency drifts upward

with amplitude along the *backbone curve* $\Omega(a)$ [8, 1]. Under harmonic forcing, the hardening resonance leans over, folds, and *jumps* between coexisting branches with hysteresis. For a resonant sensor the drift is a calibration error. For a harvester or an actuator the jump is a loss of function. In practice, the designer of such a panel is not free to move its linear frequency (set by the application) or its mass (set by the vehicle). The design question is therefore whether the *nonlinear* stiffness can be changed *independently* of both.

1.1 Related work and positioning

Reduction and design of nonlinear normal modes. The backbone is the frequency–amplitude relation of a nonlinear normal mode [1, 2, 10]. Spectral submanifold (SSM) theory makes it the reduced dynamics on the unique smoothest invariant manifold tangent to a modal subspace [3]. Its parametrization can be computed directly from finite-element models to arbitrary order [4, 5, 6, 7]; see [8] for a review of geometrically nonlinear reduction. This makes nonlinear response a *designable* quantity. Dou et al. shaped normal-form coefficients of beams and frames by adjoint shape optimization [12, 13]. Li derived explicit SSM sensitivities [14]. Pozzi et al. tailored conservative backbones through the reduced dynamics of Lyapunov subcenter manifolds [16], obtained adjoint sensitivities of SSM backbone curves at arbitrary order [15], and optimized the topology of the hardening/softening coefficient [17]. Liang et al. extended adjoint SSM sensitivities to forced-response curves and density-based topology optimization [18]. Force-based reduced-order models have also been used to optimize nonlinear finite-element models [19]. These works supply the tools we use (SSM reduction, adjoint sensitivities, gradient-based design), and we claim no novelty for them. Their design variables are the *shape*, *size* or *material density* of isotropic structures. None uses a fiber-orientation field, and none imposes the tow-curvature limit of automated fiber placement.

Variable-stiffness composites. Steering tows along curvilinear paths was introduced for buckling and stress concentration [22]. It is now a mature design space for *linear* objectives: in-plane stiffness and buckling [23, 24, 26], and fundamental frequency through lamination parameters [25, 21], including manufacturable segmentations of the optimal stiffness field [35] and layer-wise frequency optimization that models the gaps and overlaps of tow placement [34, 36], and the modal frequencies and damping of manufactured tow-steered laminates have been measured [38]. Steering has also been optimized for a geometrically nonlinear *static* objective, postbuckling stiffness [39, 40], where the gain is attributed to the in-plane redistribution of load toward the supported edges [23, 39]; the relaxed-stretch mechanism identified here is closely analogous. Orientation is also optimized pointwise in topology-optimization frameworks [47, 48], and continuous-fiber printing [49] with toolpath-integrated, curvature-constrained design [50] lays such fields directly. The *nonlinear* vibration of variable-stiffness plates has been *analyzed* for *prescribed* fiber paths: linear modes [44], forced periodic vibrations [27, 28] and free nonlinear vibrations [45, 46] (Akhavan and Ribeiro), nonlinear free vibration [29], semi-analytical parametric studies of free and forced response [30], and, most recently, backbone curves in a thermal environment with a higher-order shell element [33]. The nearest neighbor is the work of Farsadi et al. [31, 32]. They optimize curvilinear fiber paths for the *linear* fundamental frequency and then examine the amplitude-dependent frequency of the optimum.

What is new here. Relative to [31, 32], the objective is the nonlinear hardening coefficient itself, minimized *under an equality constraint on the linear frequency* at fixed mass, rather than the frequency with the nonlinear response assessed afterwards. Relative to the SSM-optimization

literature, the design variable is a fiber-orientation field under the pointwise tow-curvature limit, with the material and its mass fixed. The central contribution is not the optimizer but a *mechanism*: two strain fields of one mode weight the same anisotropic stiffness differently. A local Fourier theory then predicts when steering must help, where the fibers go and what limits the gain. Each prediction is tested against the competing laminate freedoms, a uniform angle and the stacking sequence, by comparing the attainable (frequency, hardening) sets.

Why it matters. Designers of resonant devices already exploit or cancel geometric nonlinearity by tuning design parameters, for example by balancing mechanical hardening against electrostatic softening to extend the linear range of nanomechanical sensors [20]. A composite panel offers an additional freedom that leaves geometry, mass and material unchanged.

The design quantity J, operationally. For the panel's fundamental mode with mid-plate deflection amplitude a (in meters), the backbone to cubic order is

$$\frac{\Omega(a)}{\omega_1} = 1 + \tfrac{3}{8}\, J\, a^2, \qquad J = \frac{\gamma}{\omega_1^2\, w_c^2} \quad [\mathrm{m}^{-2}],$$

where γ is the cubic coefficient of the single-mode reduced dynamics $\ddot{q} + \omega_1^2 q + \gamma q^3 = 0$ and w_c the mode's mid-plate value (Sec. 2). J does not depend on the mode normalization, and $\frac{3}{8}Ja^2$ is the relative frequency shift that a ring-down or a stepped-sine test measures at amplitude a. A ratio $J/J_{\mathrm{UD}} = 0.37$ means that, at the same amplitude, the frequency drifts 37% as much as in the unidirectional (UD) panel. Equivalently, the same drift is reached at $(0.37)^{-1/2} \approx 1.64$ times the amplitude. For the UD plate studied here ($200 \times 150 \times 2\,\mathrm{mm}$, simply supported), $f_1 = 256.3\,\mathrm{Hz}$ and the shift is 10.95% at $a = 0.4h = 0.8\,\mathrm{mm}$.

The laminate idealization. The plate designs are a single steered layer of thickness $h = 2\,\mathrm{mm}$ with one fiber angle $\theta(\boldsymbol{x})$ through the thickness at each point. Such a layer is symmetric about its mid-plane in the laminate sense: its bending–extension coupling vanishes ($\boldsymbol{B} = \boldsymbol{0}$) and its bending stiffness is $\boldsymbol{D} = \frac{h^2}{12}\boldsymbol{A}$ pointwise. It is not *balanced*, since a single angle $\theta \neq 0°, 90°$ couples extension and shear ($A_{16} \neq 0$). In practice the same in-plane and bending stiffnesses are obtained from several plies laid at the same local angle. The idealization assumes constant thickness, full coverage and no gaps or overlaps; Sec. 5.7 quantifies what tow placement would change. We call the designs *curvature-feasible idealized tow fields*: they satisfy the tow-curvature limit pointwise, which is necessary for laying them but not sufficient. Uniform stacks (Sec. 5.2) are symmetric, $\boldsymbol{B} = \boldsymbol{0}$, but in general unbalanced.

1.2 Question, hypothesis and predictions

Scientific question.

> *For a thin composite panel of fixed mass and fixed linear frequency, can the* spatial distribution *of fiber orientation reduce the geometric hardening of its fundamental mode? If so, what physical mechanism provides the lever, what bounds it, and how does it compare with the classical laminate freedoms — a uniform ply angle and the stacking sequence?*

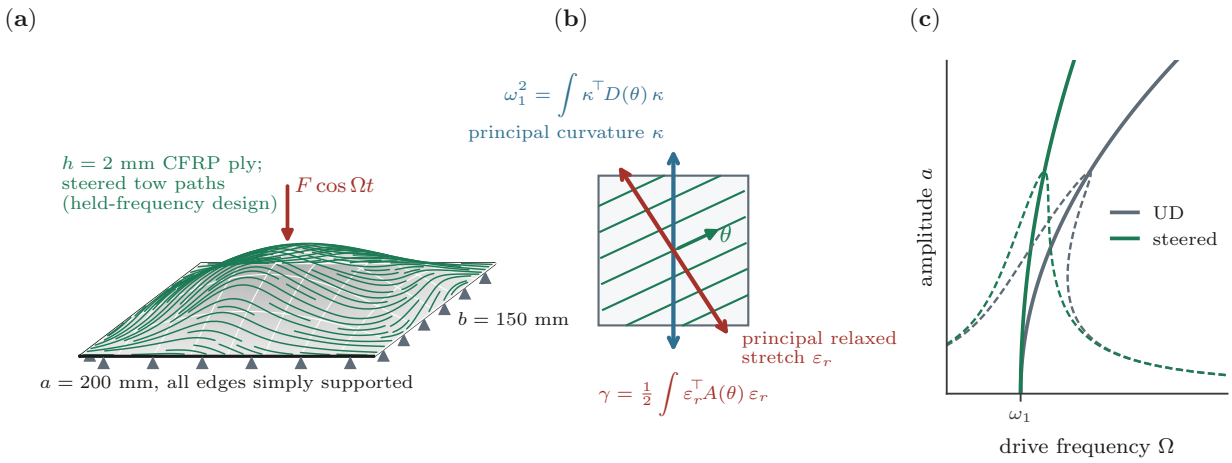


Figure 1: **One fiber angle per point must serve two strain fields of the same mode: the curvature, which sets the frequency, and the relaxed stretch, which sets the hardening.** **(a)** A simply supported $200 \times 150 \times 2$ mm CFRP plate, drawn deflected in its fundamental mode, with the tow paths of the held-frequency steered design of Sec. 5.2 and the harmonic center load. **(b)** What one material point sees: the fiber direction θ, the principal curvature of the mode, which with the bending stiffness $\boldsymbol{D}(\theta)$ sets ω_1^2, and the principal relaxed stretch (drawn at the plate's typical 33° conflict with the curvature, Sec. 5.3), which with the membrane stiffness $\boldsymbol{A}(\theta)$ sets the hardening γ (Sec. 2). **(c)** The consequence, sketched: at the same linear frequency ω_1, a smaller hardening bends the backbone (solid) less, and the forced response (dashed) of the unidirectional plate folds and jumps where the steered one stays single-valued (Sec. 5.6).

Hypothesis. (Fig. 1.) The linear frequency and the hardening are integrals of the *same* rotated stiffness tensor against two *different* strain fields of the fundamental mode. The frequency is set by the bending curvature, $\omega_1^2 = \int \boldsymbol{\kappa}^\top \boldsymbol{D}(\theta)\boldsymbol{\kappa}$. The hardening is set by the membrane stretch the mode induces after the in-plane field has relaxed, $\gamma = \frac{1}{2} \int \boldsymbol{\varepsilon}_r^\top \boldsymbol{A}(\theta)\boldsymbol{\varepsilon}_r$ (Sec. 2). The two strain fields differ from point to point in magnitude and in principal direction. We hypothesize that a spatially varying fiber angle can therefore satisfy, *pointwise*, a trade-off between the two energies that a uniform angle can only satisfy *on average*. Where stretching energy dominates, the fibers turn across the principal stretch. Where bending energy dominates, they are set toward or away from the principal curvature as the frequency constraint requires (away from it when the frequency is held at the UD value, the lowest of all uniform layups). Stacking separates the same two functionals in the classical way, through the thickness ($\boldsymbol{D}$ weights the outer plies, $\boldsymbol{A}$ all plies equally [21]); it is the competing freedom against which the in-plane route is tested. Section 3 turns the hypothesis into precise statements, which make five falsifiable predictions:

P1 An isotropic material offers no orientation lever at all (Corollary 2).

P2 For a single-ply beam, any uniform angle leaves the frequency-normalized hardening *exactly* unchanged, so every gain must come from spatial variation. The optimal variation places compliant fibers where the mode does not bend: the inflection points (Proposition 6).

P3 On a plate, a steered field reaches combinations of linear frequency and hardening that no uniform layup reaches, whether a single angle or a stacking sequence.

P4 The optimized orientation follows the local preferred directions of the two energies. A first-

order pointwise design computed from the unidirectional mode alone already captures most of the gain (Proposition 5).

P5 The lever grows with the material anisotropy and with the conflict between the two strain fields, and is limited by the tow-curvature constraint relative to the length scale of those fields.

A consequence follows if the predictions hold. The jump-free amplitude scales as $J^{-1/2}$ (Proposition 3), so reducing the hardening at held frequency widens the panel's hysteresis-free operating range by a predictable factor. That factor is a property of the conservative backbone; the absolute thresholds also depend on damping and forcing.

1.3 Approach

We test the predictions on a clamped laminated beam and on simply supported and clamped laminated plates, all von Kármán. The backbone is computed by an SSM reduction verified against analytic, harmonic-balance (EPMC) and finite-element references. For design we use the SSM's hardening coefficient in a closed form that is exact for the in-plane relaxation and evaluated on a *mode-adaptive* fundamental, with an analytic adjoint. The plate frequency is an equality constraint and the beam frequency a floor ($\omega_1 \geq 0.98\,\omega_{1,\mathrm{UD}}$). The tow-curvature limit is imposed pointwise. The held-frequency plate design is then checked on refined bases, on an independent finite-element model with the frequency re-matched, on the full multi-mode model (by its SSM, by EPMC continuation and by time integration of up/down frequency sweeps), for other supports, aspect ratios, damping laws and edge restraints, and under an as-built tow-defect model.

What "exact" means in this paper. Three levels are kept apart. *(i) Exact statements* are proved for the stated model class: the Fourier structure (Proposition 4), isotropic annihilation (Corollary 2), beam invariance (Corollary 3) and the UD floor among uniform laminates (Corollary 1). *(ii) Exact within the discretized design model*: the in-plane relaxation is exact for the adopted von Kármán kinematics and in-plane basis (Proposition 1), and the frequency equality holds to 10^{-10} on the design basis. *(iii) Approximate across models*: on finer bases and an independent finite-element model, the steered frequency differs from the UD value by up to 1.4%. We therefore also report the hardening ratio with the frequency re-matched in each of those models (Sec. 5.5). The cubic backbone itself is a small-amplitude approximation, valid over the amplitude range verified against full-order EPMC (Appendix D).

Organization. Section 2 states the model, the reduction and the backbone and jump quantities. Section 3 develops the mechanism and its predictions. Section 4 poses the design problem and derives its sensitivities. Section 5 tests P1–P5, the robustness of the result and the consequence for the forced response. Section 6 states what the results do and do not establish. The appendices give the SSM recursion, the SSM–condensation equivalence, the finite-element validation, calibration against prior work, and a verification and reproducibility map.

2 Model, reduction, and the quantities we design

2.1 Laminate constitutive model

A unidirectional CFRP lamina ($E_1 = 131\,\mathrm{GPa}$, $E_2 = 9\,\mathrm{GPa}$, $G_{12} = 5\,\mathrm{GPa}$, $\nu_{12} = 0.27$, $\rho = 1600\,\mathrm{kg/m^3}$) has the on-axis plane-stress stiffness $\boldsymbol{Q}$, with $Q_{11} = E_1/(1-\nu_{12}\nu_{21})$, $Q_{22} = E_2/(1-\nu_{12}\nu_{21})$, $Q_{12} = \nu_{12}Q_{22}$, $Q_{66} = G_{12}$ and $\nu_{21} = \nu_{12}E_2/E_1$. In engineering Voigt notation (strain

$\boldsymbol{s} = [s_{xx}, s_{yy}, 2s_{xy}]^\top$), rotating the fibers through θ gives

$$\bar{\boldsymbol{Q}}(\theta) = \boldsymbol{T}(\theta)\,\boldsymbol{Q}\,\boldsymbol{T}(\theta)^\top, \qquad \boldsymbol{T}(\theta) = \begin{bmatrix} c^2 & s^2 & -2sc \\ s^2 & c^2 & 2sc \\ sc & -sc & c^2 - s^2 \end{bmatrix}, \quad (c,s) = (\cos\theta, \sin\theta). \tag{1}$$

Its entries are trigonometric polynomials of degree four in θ (Tsai–Pagano [21]):

$$\bar{\boldsymbol{Q}}(\theta) = \boldsymbol{\Gamma}_0 + \cos 2\theta\,\boldsymbol{\Gamma}_1 + \sin 2\theta\,\boldsymbol{\Gamma}_2 + \cos 4\theta\,\boldsymbol{\Gamma}_3 + \sin 4\theta\,\boldsymbol{\Gamma}_4, \tag{2}$$

$$\boldsymbol{\Gamma}_0 = \begin{bmatrix} U_1 & U_4 & 0 \\ U_4 & U_1 & 0 \\ 0 & 0 & U_5 \end{bmatrix}, \quad \boldsymbol{\Gamma}_1 = \begin{bmatrix} U_2 & 0 & 0 \\ 0 & -U_2 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \boldsymbol{\Gamma}_2 = \tfrac{U_2}{2}\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix},$$
$$\boldsymbol{\Gamma}_3 = U_3 \begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \quad \boldsymbol{\Gamma}_4 = U_3 \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & -1 \\ 1 & -1 & 0 \end{bmatrix},$$

with the material invariants $U_1 = \frac{1}{8}(3Q_{11} + 3Q_{22} + 2Q_{12} + 4Q_{66})$, $U_2 = \frac{1}{2}(Q_{11} - Q_{22})$, $U_3 = \frac{1}{8}(Q_{11} + Q_{22} - 2Q_{12} - 4Q_{66})$, $U_4 = \frac{1}{8}(Q_{11} + Q_{22} + 6Q_{12} - 4Q_{66})$, $U_5 = \frac{1}{8}(Q_{11} + Q_{22} - 2Q_{12} + 4Q_{66})$. $\boldsymbol{\Gamma}_0$ is isotropic. All the dependence on orientation is carried by U_2 and U_3, which vanish for an isotropic material. For the CFRP above, $U_2/U_1 = 1.10$ and $U_3/U_1 = 0.26$.

A symmetric laminate ($\boldsymbol{B} = \boldsymbol{0}$) of thickness h has the membrane and bending stiffnesses

$$\boldsymbol{A} = h\Big[\boldsymbol{\Gamma}_0 + \textstyle\sum_{i=1}^{4} \xi_i^A \boldsymbol{\Gamma}_i\Big], \qquad \boldsymbol{D} = \tfrac{h^3}{12}\Big[\boldsymbol{\Gamma}_0 + \textstyle\sum_{i=1}^{4} \xi_i^D \boldsymbol{\Gamma}_i\Big], \tag{3}$$

with the lamination parameters [21, 25]

$$\boldsymbol{\xi}^A = \int_{-1/2}^{1/2} \boldsymbol{v}(\theta(\bar z))\,\mathrm{d}\bar z, \qquad \boldsymbol{\xi}^D = 12\int_{-1/2}^{1/2} \bar z^2\,\boldsymbol{v}(\theta(\bar z))\,\mathrm{d}\bar z, \qquad \boldsymbol{v}(\theta) = (\cos 2\theta, \sin 2\theta, \cos 4\theta, \sin 4\theta), \tag{4}$$

where $\theta(\bar z)$ is the ply angle at normalized height $\bar z = z/h$. Two facts organize everything that follows.

- A *single-ply-like* laminate (one angle through the thickness) has $\boldsymbol{\xi}^A = \boldsymbol{\xi}^D = \boldsymbol{v}(\theta)$, so $\boldsymbol{D} = \frac{h^2}{12}\boldsymbol{A}$ pointwise.
- A *stacking sequence* makes $\boldsymbol{\xi}^D \neq \boldsymbol{\xi}^A$. Because $\boldsymbol{\xi}^D$ weights the outer plies by $12\bar z^2$, the stacking sequence separates bending from membrane stiffness *through the thickness*.

A *steered* laminate makes $\boldsymbol{\xi}^{A,D}$ functions of the in-plane position $\boldsymbol{x}$.

2.2 The laminated von Kármán plate

A rectangular plate $\Omega = [0,a] \times [0,b]$ ($a = 0.20\,\mathrm{m}$, $b = 0.15\,\mathrm{m}$, $h = 2\,\mathrm{mm}$) carries the transverse deflection w and in-plane displacements $\boldsymbol{u} = (u, v)$. The von Kármán membrane strain and curvature are

$$\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}_L(\boldsymbol{u}) + \boldsymbol{\varepsilon}_{NL}(w), \quad \boldsymbol{\varepsilon}_L = \begin{bmatrix} u_{,x} \\ v_{,y} \\ u_{,y} + v_{,x} \end{bmatrix}, \quad \boldsymbol{\varepsilon}_{NL} = \tfrac{1}{2}\begin{bmatrix} w_{,x}^2 \\ w_{,y}^2 \\ 2w_{,x}w_{,y} \end{bmatrix}, \qquad \boldsymbol{\kappa} = -\begin{bmatrix} w_{,xx} \\ w_{,yy} \\ 2w_{,xy} \end{bmatrix}, \tag{5}$$

Table 1: Convergence of the evaluator on the reported held-frequency steered design (generated by `conv_table.py`). J/J_{UD} and the relaxed share are converged on the bold basis; three in-plane modes per direction under-resolve the steered relaxation. The steered frequency is a Ritz upper bound that falls by 0.9% as the transverse basis grows (the UD frequency does not change), so "held frequency" holds to about 0.9% across bases and 1.4% against the independent FE model. Bold: the basis used throughout.

transverse $m, n \leq$	in-plane $i, j \leq$	$f_{1,\mathrm{UD}}$ (Hz)	f_1 (Hz)	J/J_{UD}	relaxed share (UD)	relaxed share (steered)
8	3	256.3	256.3	0.523	33%	31%
8	4	256.3	256.3	0.420	33%	44%
8	5	256.3	256.3	0.375	33%	50%
8	**6**	**256.3**	**256.3**	**0.366**	**33%**	**51%**
8	8	256.3	256.3	0.360	33%	52%
12	6	256.3	254.6	0.370	33%	51%
12	8	256.3	254.6	0.364	33%	52%
16	8	256.3	253.9	0.366	33%	51%

and, with $\boldsymbol{A}(\boldsymbol{x}), \boldsymbol{D}(\boldsymbol{x})$ from (3), the strain energy is

$$U = \tfrac{1}{2}\int_\Omega \boldsymbol{\kappa}^\top \boldsymbol{D}\,\boldsymbol{\kappa}\,\mathrm{d}\Omega + \tfrac{1}{2}\int_\Omega (\boldsymbol{\varepsilon}_L + \boldsymbol{\varepsilon}_{NL})^\top \boldsymbol{A}\,(\boldsymbol{\varepsilon}_L + \boldsymbol{\varepsilon}_{NL})\,\mathrm{d}\Omega. \tag{6}$$

With the kinetic energy $T = \frac{1}{2}\int \rho h(\dot{w}^2 + |\dot{\boldsymbol{u}}|^2)$, Hamilton's principle gives the laminated Föppl–von Kármán equations [11],

$$N_{\alpha\beta,\beta} = \rho h\,\ddot{u}_\alpha, \qquad M_{\alpha\beta,\alpha\beta} + \left(N_{\alpha\beta} w_{,\beta}\right)_{,\alpha} + p = \rho h\,\ddot{w}, \qquad \boldsymbol{N} = \boldsymbol{A}(\boldsymbol{\varepsilon}_L + \boldsymbol{\varepsilon}_{NL}),\ \boldsymbol{M} = \boldsymbol{D}\boldsymbol{\kappa}. \tag{7}$$

The edges are simply supported and immovable ($w = 0$, $\boldsymbol{u} = \boldsymbol{0}$ on $\partial\Omega$). We discretize by Galerkin's method on sine bases, which satisfy these conditions term by term:

$$w = \sum_{k=1}^{N_w} q_k\,\phi_k, \quad \phi_k = \sin\tfrac{m_k \pi x}{a}\sin\tfrac{n_k \pi y}{b}; \qquad u, v = \sum_{j=1}^{N_s} s_j^{u,v}\,\psi_j, \quad \psi_j = \sin\tfrac{i_j \pi x}{a}\sin\tfrac{l_j \pi y}{b}. \tag{8}$$

We use $N_w = 64$ ($m, n \leq 8$) and $N_s = 36$ per component. For the robustness study of Sec. 5.7 the same evaluator is built on two other bases: clamped edges ($w = \partial_n w = 0$) with $\phi_k = \sin\frac{\pi x}{a}\sin\frac{m_k \pi x}{a}\sin\frac{\pi y}{b}\sin\frac{n_k \pi y}{b}$, which reproduces the clamped isotropic square plate's frequency parameter 35.99 to 0.15%; and in-plane traction-free ("movable") edges, with a Chebyshev polynomial basis of degree ≤ 10 for u, v, converged to four digits in γ. The bases are converged for the steered designs: Table 1 shows that the steered relaxation needs at least five to six in-plane modes per direction; the three-mode basis that suffices for a uniform laminate under-resolves it badly ($J/J_{\mathrm{UD}} = 0.52$ instead of 0.37).

With $\boldsymbol{x} = [\boldsymbol{q}; \boldsymbol{s}]$ the system is

$$\boldsymbol{M}\ddot{\boldsymbol{x}} + \boldsymbol{C}\dot{\boldsymbol{x}} + \boldsymbol{K}\boldsymbol{x} + \boldsymbol{B}_2(\boldsymbol{x}, \boldsymbol{x}) + \boldsymbol{T}_3(\boldsymbol{x}, \boldsymbol{x}, \boldsymbol{x}) = \boldsymbol{f}(t), \tag{9}$$

with $M_{kl} = \rho h \int \phi_k \phi_l$ (likewise for ψ), $\boldsymbol{K} = \mathrm{blkdiag}(\boldsymbol{K}_b, \boldsymbol{K}_{\mathrm{in}})$, $(\boldsymbol{K}_b)_{kl} = \int \boldsymbol{\kappa}_k^\top \boldsymbol{D}\boldsymbol{\kappa}_l$ and $(\boldsymbol{K}_{\mathrm{in}})_{jl} = \int \boldsymbol{\varepsilon}_{L,j}^\top \boldsymbol{A}\boldsymbol{\varepsilon}_{L,l}$. $\boldsymbol{B}_2$ is the gradient of $\int \boldsymbol{\varepsilon}_L^\top \boldsymbol{A}\boldsymbol{\varepsilon}_{NL}$ (the $\boldsymbol{u} \leftrightarrow w^2$ coupling) and $\boldsymbol{T}_3$ is the gradient of $\frac{1}{2}\int \boldsymbol{\varepsilon}_{NL}^\top \boldsymbol{A}\boldsymbol{\varepsilon}_{NL}$ (the membrane cubic). Because $\boldsymbol{B} = \boldsymbol{0}$, the *linear* problem decouples into a bending block and an in-plane block. The mass is design-independent: every design in this paper has the same plate, the same material and the same thickness.

The *beam* is the one-dimensional specialization: a clamped–clamped strip ($L = 0.30\,\mathrm{m}$, width $20\,\mathrm{mm}$, $h = 2\,\mathrm{mm}$) with axial u and transverse w, $\varepsilon = u' + \frac{1}{2}w'^2$ and $\kappa = -w''$. For a single-ply laminate, $A = bhE(x)$ and $D = \frac{bh^3}{12}E(x)$ with $E(x) = \bar{Q}_{11}(\theta(x))$. It is discretized by Hermite-cubic/linear elements, and the axial ends are immovable.

2.3 Spectral submanifold and backbone

In first-order form $\boldsymbol{z} = [\boldsymbol{x}; \dot{\boldsymbol{x}}]$, $\boldsymbol{\mathcal{B}}\dot{\boldsymbol{z}} = \boldsymbol{\mathcal{A}}\boldsymbol{z} + \boldsymbol{F}(\boldsymbol{z})$ with $\boldsymbol{\mathcal{B}} = \left[\begin{smallmatrix} \boldsymbol{C} & \boldsymbol{M} \\ \boldsymbol{M} & \boldsymbol{0} \end{smallmatrix}\right]$, $\boldsymbol{\mathcal{A}} = \left[\begin{smallmatrix} -\boldsymbol{K} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{M} \end{smallmatrix}\right]$ and $\boldsymbol{F} = [-\boldsymbol{B}_2 - \boldsymbol{T}_3; \boldsymbol{0}]$. Let $\lambda_1, \bar{\lambda}_1$ be the eigenvalues of the fundamental, with right eigenvector $\boldsymbol{v}_1$ and left eigenvector $\boldsymbol{u}_1$ ($\boldsymbol{u}_1^*\boldsymbol{\mathcal{A}} = \lambda_1\boldsymbol{u}_1^*\boldsymbol{\mathcal{B}}$, $\boldsymbol{u}_1^*\boldsymbol{\mathcal{B}}\boldsymbol{v}_1 = 1$). The SSM $\mathcal{W}$ is the unique smoothest invariant manifold tangent to their eigenspace [3]. It exists if λ_1 is the slowest pair and no outer eigenvalue equals $m_1\lambda_1 + m_2\bar{\lambda}_1$ up to the truncation order. Parametrizing $\boldsymbol{z} = \boldsymbol{W}(\boldsymbol{p})$ and $\dot{\boldsymbol{p}} = \boldsymbol{R}(\boldsymbol{p})$ gives the invariance equation $\boldsymbol{\mathcal{B}}\,\mathrm{D}\boldsymbol{W}\,\boldsymbol{R} = \boldsymbol{\mathcal{A}}\boldsymbol{W} + \boldsymbol{F}(\boldsymbol{W})$. Expanding both maps in monomials $\boldsymbol{p}^{\boldsymbol{m}}$, it separates order by order into the homological equations

$$(\Lambda_{\boldsymbol{m}}\boldsymbol{\mathcal{B}} - \boldsymbol{\mathcal{A}})\boldsymbol{w}_{\boldsymbol{m}} + \boldsymbol{\mathcal{B}}(\boldsymbol{v}_1 R^1_{\boldsymbol{m}} + \bar{\boldsymbol{v}}_1 R^2_{\boldsymbol{m}}) = \boldsymbol{G}_{\boldsymbol{m}}, \qquad \Lambda_{\boldsymbol{m}} = m_1\lambda_1 + m_2\bar{\lambda}_1, \tag{10}$$

where $\boldsymbol{G}_{\boldsymbol{m}}$ collects lower-order terms (Appendix A). At the near-resonant indices $\boldsymbol{m} = (k{+}1, k)$ the operator is (near-)singular. Projecting onto $\boldsymbol{u}_1$ gives the reduced (normal-form) coefficients $\gamma_{2k+1} = \boldsymbol{u}_1^*\boldsymbol{G}_{(k+1,k)}$. On the orbit $p_1 = \frac{\rho}{2}e^{i\vartheta}$, $p_2 = \bar{p}_1$, the backbone and the modal decay rate are

$$\Omega(\rho) = \mathrm{Im}\left(\lambda_1 + \sum_{k\geq 1}\gamma_{2k+1}\left(\tfrac{\rho}{2}\right)^{2k}\right), \qquad \alpha(\rho) = \mathrm{Re}(\cdots). \tag{11}$$

The implementation is verified against the analytic Duffing backbone through fifth order ($3/8$ and $-21/256$), against full-order EPMC continuation, and against two published results [15] (Appendix D). The master eigenvector is computed by Rayleigh-quotient inverse iteration on the complex-symmetric pencil, with $\boldsymbol{u}_1 = \bar{\boldsymbol{v}}_1$. A dense QZ solve of the stiff first-order pencil was found to lose up to ten digits (eigenvector condition number $\sim 10^{14}$ for laminated beams) and is no longer used for symmetric systems (regression test in `tests/test_relaxed_plate.py`).

2.4 The relaxed hardening coefficient

Let the fundamental be the lowest Ritz pair of the bending block, $\boldsymbol{K}_b\boldsymbol{v} = \omega_1^2\boldsymbol{M}_w\boldsymbol{v}$ with $\boldsymbol{v}^\top\boldsymbol{M}_w\boldsymbol{v} = 1$. Its shape $w_1 = \sum_k v_k\phi_k$ *adapts* to the stiffness field. This is the exact linear master mode of (9), because the blocks decouple when $\boldsymbol{B} = \boldsymbol{0}$. Write its curvature as $\boldsymbol{\kappa}_1$ and its stretch as

$$\boldsymbol{m} = \left[w_{1,x}^2,\ w_{1,y}^2,\ 2w_{1,x}w_{1,y}\right]^\top = 2\boldsymbol{\varepsilon}_{NL}(w_1). \tag{12}$$

Proposition 1 (Relaxed cubic, variational form)**.** *Let the in-plane field be slaved quasi-statically to q^2, the square of the fundamental's modal amplitude. Then the cubic coefficient of the single-mode reduced dynamics $\ddot{q} + \omega_1^2 q + \gamma q^3 = 0$ is*

$$\gamma = \min_{\boldsymbol{s}}\ \tfrac{1}{2}\int_\Omega (\boldsymbol{m} - 2\boldsymbol{\varepsilon}_L(\boldsymbol{s}))^\top \boldsymbol{A}\,(\boldsymbol{m} - 2\boldsymbol{\varepsilon}_L(\boldsymbol{s}))\,\mathrm{d}\Omega = k_{nl} - 2\,\boldsymbol{g}^\top\boldsymbol{K}_{\mathrm{in}}^{-1}\boldsymbol{g} = \tfrac{1}{2}\int_\Omega \boldsymbol{\varepsilon}_r^\top\boldsymbol{A}\,\boldsymbol{\varepsilon}_r\,\mathrm{d}\Omega, \tag{13}$$

with $k_{nl} = \frac{1}{2}\int \boldsymbol{m}^\top\boldsymbol{A}\boldsymbol{m}$, $\boldsymbol{g} = \frac{1}{2}\int \boldsymbol{\varepsilon}_L(\cdot)^\top\boldsymbol{A}\boldsymbol{m}$, $\boldsymbol{s}^\star = \boldsymbol{K}_{\mathrm{in}}^{-1}\boldsymbol{g}$ and the relaxed stretch $\boldsymbol{\varepsilon}_r = \boldsymbol{m} - 2\boldsymbol{\varepsilon}_L(\boldsymbol{s}^\star)$.

Proof. With $w = qw_1$ and $\boldsymbol{u} = -q^2\boldsymbol{s}$, the membrane strain is $\boldsymbol{\varepsilon} = \frac{q^2}{2}(\boldsymbol{m} - 2\boldsymbol{\varepsilon}_L(\boldsymbol{s}))$. The membrane energy is therefore $\frac{q^4}{8}\int(\boldsymbol{m} - 2\boldsymbol{\varepsilon}_L(\boldsymbol{s}))^\top\boldsymbol{A}(\cdot) = \frac{q^4}{4}F(\boldsymbol{s})$, where $F(\boldsymbol{s}) = k_{nl} - 4\boldsymbol{s}^\top\boldsymbol{g} + 2\boldsymbol{s}^\top\boldsymbol{K}_{\mathrm{in}}\boldsymbol{s}$. Static equilibrium of the in-plane field minimizes F, at $\boldsymbol{s}^\star = \boldsymbol{K}_{\mathrm{in}}^{-1}\boldsymbol{g}$, with $F(\boldsymbol{s}^\star) = k_{nl} - 2\boldsymbol{g}^\top\boldsymbol{K}_{\mathrm{in}}^{-1}\boldsymbol{g}$. The reduced potential $\frac{\gamma}{4}q^4$ then gives (13). The last form is $F(\boldsymbol{s}^\star)$ written out. □

Proposition 1 has three uses. (i) k_{nl} is the restrained (immovable in-plane field) upper bound, and $2\boldsymbol{g}^\top \boldsymbol{K}_{\text{in}}^{-1}\boldsymbol{g}$ is the *relaxation*. Its share of k_{nl} is 33% for the unidirectional plate. (ii) Because γ is a *minimum* over $\boldsymbol{s}$, its design derivative needs no in-plane adjoint (envelope theorem; Sec. 4.3). (iii) It writes the hardening as the integral of an energy density (Sec. 3).

Proposition 2 (Agreement with the SSM)**.** *For an undamped system* (9) *with* $\boldsymbol{B} = \boldsymbol{0}$*, the cubic SSM coefficient of the fundamental satisfies* $\operatorname{Im}\gamma_3 = \frac{3}{2\omega_1}\gamma\,[1 + O(4\omega_1^2/\omega_{\text{in}}^2)]$*, where* ω_{in} *is the lowest in-plane frequency.*

The proof is in Appendix B. In short: the $(1,1)$ in-plane homological solve is exactly static ($\Lambda_{(1,1)} = 0$), and the $(2,0)$ solve is static up to the resolvent factor $(\boldsymbol{K}_{\text{in}} - 4\omega_1^2\boldsymbol{M}_{\text{in}})^{-1}$. For our plate $4\omega_1^2/\omega_{\text{in}}^2 = 3.2\times10^{-3}$ ($f_{\text{in}} \approx 9.0\,\text{kHz}$ on the full multi-mode plate), and (13) reproduces the full multi-mode SSM coefficient to within 1.2% (Sec. 5.5).

2.5 The backbone in physical amplitude, and the design quantity J

For $\ddot{q} + \omega_1^2 q + \gamma q^3 = 0$ the cubic SSM gives $\Omega = \omega_1 + \frac{3}{8}(\gamma/\omega_1)q^2$. The mid-plate deflection of the mode is $a = w_c q$, with $w_c = w_1(a/2, b/2)$, so

$$\frac{\Omega}{\omega_1} = 1 + \frac{3}{8}J\,a^2 = 1 + \beta\Big(\frac{a}{h}\Big)^2, \qquad J = \frac{\gamma}{\omega_1^2 w_c^2}, \qquad \beta = \tfrac{3}{8}Jh^2. \tag{14}$$

J (units m^{-2}) is the *frequency-normalized detuning*. It is invariant to the mode normalization, and it is what we design. The unidirectional plate has $f_1 = 256.3\,\text{Hz}$, $\beta = 0.684$ and a 10.95% shift at $a/h = 0.4$. The cubic truncation reproduces the full-order EPMC detuning to within 5% over the amplitudes EPMC reached, $a/h \le 0.21$ (UD) and ≤ 0.35 (steered) (Appendix D).

2.6 Forced response and the jump-free amplitude

With harmonic forcing $\boldsymbol{f}\cos\Omega t$ and damping $\boldsymbol{C} = 2\zeta\omega_1\boldsymbol{M}$, seeking $p_1 = \frac{\rho}{2}e^{i\Omega t}$ on the SSM gives the leading-order forced-response curve

$$\big[(\Omega - \Omega(\rho))^2 + \alpha(\rho)^2\big]\rho^2 = |f_1|^2, \qquad f_1 = \boldsymbol{u}_1^*[\boldsymbol{f};\boldsymbol{0}]. \tag{15}$$

Proposition 3 (Jump-free amplitude)**.** *For the cubic backbone* (14) *with* $\alpha = \zeta\omega_1$*, the forced response* (15) *is single-valued in* Ω *for every drive whose peak (resonant) amplitude is below*

$$\frac{a_c}{h} = \Big(\frac{8\zeta}{3\sqrt{3}\,\beta}\Big)^{1/2} \;\propto\; J^{-1/2}, \tag{16}$$

and multivalued (a fold pair with jump and hysteresis) above it. At the critical drive the vertical tangent sits at the lower amplitude $(\sqrt{3}/2)\,a_c$ *on the response curve.*

Proof. Write $x = a^2$, $c = \omega_1\beta/h^2$, $\mu = \zeta\omega_1$, and let F be the drive expressed in physical amplitude. Then (15) reads $[(\sigma - cx)^2 + \mu^2]x = F^2$, with $\sigma = \Omega - \omega_1$. Put $\sigma = cx + s$. The curve is $x = F^2/(s^2+\mu^2)$, $\sigma = cF^2/(s^2+\mu^2)+s$, and it folds where $\mathrm{d}\sigma/\mathrm{d}s = 0$, i.e. where $2cF^2 s = (s^2+\mu^2)^2$. A fold pair first appears when this equation has a double root, i.e. when $2cF^2 = 4s(s^2+\mu^2)$ also holds. Combining the two conditions gives $s^2 + \mu^2 = 4s^2$, so $s = \mu/\sqrt{3}$, and the critical drive is $F_c^2 = (s^2+\mu^2)^2/(2cs) = 8\mu^3/(3\sqrt{3}\,c)$. The vertical tangent lies at $x = F_c^2/(s^2+\mu^2) = 2\mu/(\sqrt{3}c)$, while the peak of the response (at $s = 0$) is $x_c = F_c^2/\mu^2 = 8\mu/(3\sqrt{3}\,c)$, which is (16). Below F_c the curve has no fold; above it, it has two. □

Proposition 3 is the consequence we test. Holding ω_1, ζ and h and reducing J by a factor r widens the jump-free amplitude by $r^{-1/2}$. In Sec. 5.5 we compute a_c *numerically*, as the drive at which the full-order damped SSM response first becomes multivalued, and compare it with (16).

3 Theory: why a fiber field can decouple hardening from frequency

3.1 Two energy densities over one stiffness

With the mass-normalized adapted mode, the Rayleigh quotient and Proposition 1 write both design quantities as integrals of local energy densities:

$$\omega_1^2 = \int_\Omega e_K \,\mathrm{d}\Omega, \quad e_K = \boldsymbol{\kappa}_1^\top \boldsymbol{D}\, \boldsymbol{\kappa}_1; \qquad \gamma = \int_\Omega e_G \,\mathrm{d}\Omega, \quad e_G = \tfrac{1}{2}\boldsymbol{\varepsilon}_r^\top \boldsymbol{A}\, \boldsymbol{\varepsilon}_r. \tag{17}$$

$J = \gamma/(\omega_1^2 w_c^2)$ is their ratio up to the mode's mid-plate value. Both densities read the *same* rotated material (2), but through different strain fields: $\boldsymbol{\kappa}_1$ (second derivatives of w_1, largest where the mode bends) and $\boldsymbol{\varepsilon}_r$ (squared first derivatives minus the in-plane relief). For the simply supported fundamental both peak at the center, but with different principal directions (Sec. 5.3). The following proposition shows exactly how each density depends on the local fiber angle.

3.2 Local Fourier structure

Proposition 4 (Angular dependence of a laminate energy density)**.** *Let $\boldsymbol{s}$ be an engineering strain vector whose tensor has principal values $e_{\max}, e_{\min}$, ordered so that $|e_{\max}| \geq |e_{\min}|$, and let $\hat{\varphi}$ be the direction of $e_{\max}$. Then for the single-ply stiffness* (2)*,*

$$\boldsymbol{s}^\top \bar{\boldsymbol{Q}}(\theta)\boldsymbol{s} = \underbrace{U_1(e_{\max}^2 + e_{\min}^2) + 2U_4 e_{\max} e_{\min}}_{isotropic} + U_2\,\Delta\cos 2(\theta - \hat{\varphi}) + U_3\,\Sigma\cos 4(\theta - \hat{\varphi}), \tag{18}$$

with $\Delta = e_{\max}^2 - e_{\min}^2 \geq 0$ and $\Sigma = (e_{\max} - e_{\min})^2 \geq 0$.

Proof. Rotation invariance gives $\boldsymbol{s}^\top \bar{\boldsymbol{Q}}(\theta)\boldsymbol{s} = \boldsymbol{s}'^\top \bar{\boldsymbol{Q}}(\theta - \hat{\varphi})\boldsymbol{s}'$, with $\boldsymbol{s}' = [e_{\max}, e_{\min}, 0]^\top$ the strain in its principal frame. Let $\psi = \theta - \hat{\varphi}$. Using (2), $\bar{Q}_{11} = U_1 + U_2\cos 2\psi + U_3\cos 4\psi$, $\bar{Q}_{22} = U_1 - U_2\cos 2\psi + U_3\cos 4\psi$ and $\bar{Q}_{12} = U_4 - U_3\cos 4\psi$, we get $\bar{Q}_{11}e_{\max}^2 + 2\bar{Q}_{12}e_{\max}e_{\min} + \bar{Q}_{22}e_{\min}^2 = U_1(e_{\max}^2 + e_{\min}^2) + 2U_4 e_{\max}e_{\min} + U_2(e_{\max}^2 - e_{\min}^2)\cos 2\psi + U_3(e_{\max} - e_{\min})^2\cos 4\psi$. □

Applying (18) to the two densities, pointwise:

$$e_K(\boldsymbol{x}, \theta) = \tfrac{h^3}{12}\big[I_K + U_2\Delta_K\cos 2(\theta - \varphi_K) + U_3\Sigma_K\cos 4(\theta - \varphi_K)\big], \tag{19}$$

$$e_G(\boldsymbol{x}, \theta) = \tfrac{h}{2}\big[I_G + U_2\Delta_G\cos 2(\theta - \varphi_G) + U_3\Sigma_G\cos 4(\theta - \varphi_G)\big], \tag{20}$$

where $(\Delta_K, \Sigma_K, \varphi_K)$ come from the principal curvatures of w_1 and $(\Delta_G, \Sigma_G, \varphi_G)$ from the principal relaxed stretches. For CFRP $U_2 \approx 4.2\,U_3$, so the 2θ harmonic dominates. To leading order, then:

- e_K is largest (the point contributes most to ω_1^2) when the fiber lies *along* the principal curvature, $\theta = \varphi_K$;
- e_G is smallest (the point contributes least to γ) when the fiber lies *across* the principal stretch, $\theta = \varphi_G + \pi/2$.

Which way the frequency pulls depends on the sign of the multiplier Λ of Proposition 5 below, which weights e_K against e_G. If the target frequency lies above the one the stretch alone would select, $\Lambda > 0$ and the fiber is drawn toward the principal curvature, $\theta = \varphi_K$. If it lies below, $\Lambda < 0$ and the fiber is pushed *across* the principal curvature, $\theta = \varphi_K + \pi/2$, the bending-compliant direction. We call $\varphi_G + \pi/2$ and the frequency-favored direction the *preferred directions*, and the angle between them, $\chi(\boldsymbol{x}) \in [0, \pi/2]$, the *conflict*. Where $\chi = 0$, one fiber direction serves both energies. Where χ is large, any single direction compromises.

Corollary 1 (The unidirectional floor)**.** *Among all uniform symmetric laminates of the same thickness and mass, whatever the stacking sequence, the lowest linear frequency is attained by a single-angle laminate.*

Proof. For a symmetric laminate $\boldsymbol{D} = \frac{h^3}{12}\big(\boldsymbol{\Gamma}_0 + \sum_k \xi_k^D \boldsymbol{\Gamma}_k\big)$ is affine in its four lamination parameters $\boldsymbol{\xi}^D$, and the set of $\boldsymbol{\xi}^D$ reachable by stacking lies in the convex hull of the single-angle curve $\boldsymbol{\xi}^D(\theta) = (\cos 2\theta, \sin 2\theta, \cos 4\theta, \sin 4\theta)$ [21]. ω_1^2 is the minimum over admissible w of a Rayleigh quotient that is affine in $\boldsymbol{\xi}^D$, hence a concave function of $\boldsymbol{\xi}^D$, and a concave function on a convex hull attains its minimum at an extreme point, i.e. at a single-angle laminate. □

For the plate studied here the single-angle frequency rises monotonically from $\theta = 0°$ (Sec. 5.2), so UD is the lowest-frequency uniform laminate of *any* stacking sequence, not only of the enumerated ones, and at $\omega_{1,\mathrm{UD}}$ there is no other uniform competitor.

Corollary 2 (P1: isotropy annihilates the lever)**.** *If $U_2 = U_3 = 0$, then $\boldsymbol{A}$ and $\boldsymbol{D}$ are independent of every ply angle at every point. Hence ω_1, γ and J are independent of the orientation field, uniform or steered or stacked, and every orientation sensitivity vanishes identically. This is exact, including mode adaptation and relaxation, because the operators themselves do not depend on θ.*

Numerically, $\partial J/\partial\theta \equiv 0$ to 10^{-10} relative when $\boldsymbol{Q}$ is set isotropic (`tests/test_relaxed_plate.py`).

3.3 Two separation channels

Inserting (3) into (17) gives

$$e_K = \tfrac{h^3}{12}\Big[\boldsymbol{\kappa}_1^\top \boldsymbol{\Gamma}_0 \boldsymbol{\kappa}_1 + \sum_i \xi_i^D(\boldsymbol{x})\, \boldsymbol{\kappa}_1^\top \boldsymbol{\Gamma}_i \boldsymbol{\kappa}_1\Big], \qquad e_G = \tfrac{h}{2}\Big[\boldsymbol{\varepsilon}_r^\top \boldsymbol{\Gamma}_0 \boldsymbol{\varepsilon}_r + \sum_i \xi_i^A(\boldsymbol{x})\, \boldsymbol{\varepsilon}_r^\top \boldsymbol{\Gamma}_i \boldsymbol{\varepsilon}_r\Big]. \tag{21}$$

The frequency reads only the bending lamination parameters, and the hardening only the membrane ones. Each design class corresponds to a different freedom in the field $(\boldsymbol{\xi}^A(\boldsymbol{x}), \boldsymbol{\xi}^D(\boldsymbol{x}))$:

class	$\boldsymbol{\xi}^A$	$\boldsymbol{\xi}^D$	separation available
uniform single angle	$\boldsymbol{v}(\theta_0)$	$= \boldsymbol{\xi}^A$	none (one scalar)
uniform stack	const.	const. $\neq \boldsymbol{\xi}^A$	through the thickness
steered single ply	$\boldsymbol{v}(\theta(\boldsymbol{x}))$	$= \boldsymbol{\xi}^A$	in the plane
steered stack	$\mathcal{R}_{\theta(\boldsymbol{x})}\boldsymbol{\xi}_0^A$	$\mathcal{R}_{\theta(\boldsymbol{x})}\boldsymbol{\xi}_0^D$	both

Here $\mathcal{R}_\theta$ rotates the parameter pairs, $(\xi_1 + i\xi_2) \mapsto e^{2i\theta}(\xi_1 + i\xi_2)$ and $(\xi_3 + i\xi_4) \mapsto e^{4i\theta}(\xi_3 + i\xi_4)$. Section 5.2 measures the achievable (ω_1, J) set of each class.

3.4 Pointwise optimality: what an optimal field must do

Consider the problem

$$\min_{\theta(\cdot)} \; J \quad \text{s.t.} \quad \omega_1 = \omega_\star \tag{22}$$

over single-ply fields. Freeze w_1 and $\boldsymbol{\varepsilon}_r$ at a reference design. This *frozen-field* model is exact to first order in the design change: $\boldsymbol{s}^\star$ is stationary (Proposition 1) and ω_1^2 is a Rayleigh quotient. In this model w_c is fixed, so (22) reads $\min \int e_G(\boldsymbol{x}, \theta(\boldsymbol{x}))$ subject to $\int e_K(\boldsymbol{x}, \theta(\boldsymbol{x})) = \omega_\star^2$.

Proposition 5 (Pointwise optimality and strict benefit of steering)**.** *In the frozen-field model the optimal field is*

$$\theta^\star(\boldsymbol{x}) = \arg\min_{\theta} \big[e_G(\boldsymbol{x},\theta) - \Lambda\, e_K(\boldsymbol{x},\theta)\big], \tag{23}$$

with the scalar multiplier Λ fixed by the frequency constraint. A uniform angle θ_u can be optimal among all fields only if

$$U_2\Delta_G \sin 2(\theta_u-\varphi_G)+2U_3\Sigma_G \sin 4(\theta_u-\varphi_G) = \Lambda'\big[U_2\Delta_K \sin 2(\theta_u-\varphi_K)+2U_3\Sigma_K \sin 4(\theta_u-\varphi_K)\big] \tag{24}$$

holds at almost every $\boldsymbol{x}$, with one constant $\Lambda' = \frac{h^2}{6}\Lambda$. Otherwise a steered field strictly lowers J at the same frequency.

Proof. The Lagrangian $\int e_G - \Lambda(\int e_K - \omega_\star^2)$ has no derivative coupling between points, so it is minimized pointwise, giving (23). Every uniform angle is a feasible field, so the field optimum is never worse. If the uniform optimum were also the field optimum, it would satisfy the pointwise stationarity condition $\partial_\theta e_G = \Lambda \partial_\theta e_K$ at every point. By (19)–(20) this is (24). □

Remark 1 (Scope of Proposition 5). The proposition is a first-order statement about the frozen-field model. Its second part says that if (24) fails, the uniform angle is not stationary among fields, so a feasible descent direction exists and some steered field near it has a lower J at the same frequency. It does *not* say that $\theta^\star$ of (23) is the optimum of the full problem (22), in which the mode and the relaxed stretch adapt to the design and the field must be smooth and curvature-feasible. Nor does it bound how large the improvement is. Those questions are answered numerically in Sec. 5.

Condition (24) asks the anisotropic parts of two different strain fields to be proportional with one constant at a single angle. That fails whenever the conflict $\chi(\boldsymbol{x})$ or the ratio Δ_G/Δ_K varies over the plate. Since the principal curvature and the principal relaxed stretch of a bending mode turn differently over the plate, and their energy ratio varies, it fails generically. Proposition 5 is the precise form of the hypothesis. It also predicts that the lever shrinks when the conflict is small: if the two fields share their principal directions everywhere ($\chi \equiv 0$) and their ratio is nearly uniform, a single angle comes close to satisfying (24). It also yields a test (P4): $\theta^\star$ computed from the *unidirectional* fields alone, with no optimizer, should already capture much of the gain, and the optimized field should resemble it.

3.5 One dimension in closed form

Proposition 6 (Beam: harmonic against curvature-weighted mean)**.** *Consider a symmetric single-ply beam with immovable ends, $A = bhE(x)$ and $D = \frac{bh^3}{12}E(x)$, and a mass-normalized fundamental φ with mid-span value φ_c. Then*

$$\gamma = \frac{bh\,(\int_0^L \varphi'^2)^2}{2\int_0^L E^{-1}}, \qquad \omega_1^2 = \frac{bh^3}{12}\int_0^L E\,\varphi''^2, \qquad J = \frac{6}{h^2\varphi_c^2}\,\frac{(\int \varphi'^2)^2}{\int E^{-1}\int E\varphi''^2}. \tag{25}$$

Proof. Axial equilibrium makes N uniform. Integrating $\varepsilon = N/A = u' + \frac{1}{2}q^2\varphi'^2$ over the span with $u(0) = u(L) = 0$ gives $N\int A^{-1} = \frac{1}{2}q^2\int \varphi'^2$. The membrane energy is then $\frac{1}{2}N^2\int A^{-1} = q^4(\int \varphi'^2)^2/(8\int A^{-1})$, which equals $\frac{\gamma}{4}q^4$. The frequency is the Rayleigh quotient, and J follows from (14). □

Corollary 3 (P2: uniform invariance)**.** *For uniform $E \equiv E_0$, $\int E^{-1}\int E\varphi''^2 = L\int \varphi''^2$, and φ itself is independent of E_0 (the operators scale together). Hence J is* exactly *independent of any uniform single-ply angle, and every change in J is caused by spatial variation of $E(x)$.*

Equation (25) makes the mechanism transparent. The hardening sees the *harmonic* mean of E over the whole span, which is dominated by the softest segment. The frequency sees the mean of E *weighted by the curvature* φ''^2, which vanishes at the inflection points ($x/L \approx 0.224$ and 0.776 for the clamped fundamental). Compliant fibers placed there raise $\int E^{-1}$, and so lower the hardening, at almost no cost in frequency.

Corollary 4 (Bang-bang bound)**.** *Suppose $E = E_{\max}$ except on zones of total length ℓ where $E = E_{\min}$ and $\varphi'' \approx 0$. Then, to frozen-mode accuracy,*

$$J/J_{\mathrm{UD}} \approx \big[1 + (\ell/L)(E_{\max}/E_{\min} - 1)\big]^{-1}. \tag{26}$$

For CFRP, $E_{\max}/E_{\min} = \bar{Q}_{11}(0)/\bar{Q}_{11}(90^\circ) = 14.6$, so $\ell/L = 0.05$ already gives $J/J_{\mathrm{UD}} \approx 0.59$. Equation (26) contains P5 in one dimension. The lever grows with the anisotropy ratio and with how narrow the compliant zones can be made while staying where $\varphi'' \approx 0$. The narrowness is set by the tow curvature: turning a tow by 90° at rate κ_{tow} needs a run of $(\pi/2)/\kappa_{\mathrm{tow}}$, which is 0.10 m at 900°/m, a third of the span.

4 Design problems and sensitivities

4.1 Orientation field and the tow-curvature constraint

The fiber angle is a smooth field interpolated from M support values $\hat{\boldsymbol{\theta}}$ by normalized compactly supported Wendland C^2 radial basis functions [52],

$$\theta(\boldsymbol{x}) = \sum_{i=1}^{M} \Phi_i(\boldsymbol{x})\hat{\theta}_i, \qquad \Phi_i = \frac{\phi(\|\boldsymbol{x} - \boldsymbol{x}_i\|/r_s)}{\sum_j \phi(\|\boldsymbol{x} - \boldsymbol{x}_j\|/r_s)}, \qquad \phi(q) = (1-q)_+^4(4q+1), \tag{27}$$

on a regular support grid of spacing h_s with $r_s = 1.5h_s$. The baseline is $h_s = 50$ mm ($M = 20$); Sec. 5.4 refines it. The spatial derivatives $\boldsymbol{\Phi}_x, \boldsymbol{\Phi}_y$ are analytic. A tow laid along $\boldsymbol{n} = (\cos\theta, \sin\theta)$ turns at the rate

$$\kappa_{\mathrm{tow}}(\boldsymbol{x}) = (\boldsymbol{n}\cdot\nabla)\theta = \cos\theta\,\theta_{,x} + \sin\theta\,\theta_{,y}, \tag{28}$$

so its steering radius is $1/|\kappa_{\mathrm{tow}}|$. AFP and continuous-fiber printing require $|\kappa_{\mathrm{tow}}| \le \zeta_{\mathrm{all}} = 1/R_{\min}$; this pointwise form follows [50]. We impose this at every quadrature point e as $c_e = 1 - \kappa_{\mathrm{tow},e}^2/\zeta_{\mathrm{all}}^2 \ge 0$, with the analytic Jacobian

$$\frac{\partial \kappa_{\mathrm{tow},e}}{\partial \hat{\boldsymbol{\theta}}} = \big(-\sin\theta_e\,\boldsymbol{\Phi}_{x,e}\hat{\boldsymbol{\theta}} + \cos\theta_e\,\boldsymbol{\Phi}_{y,e}\hat{\boldsymbol{\theta}}\big)\boldsymbol{\Phi}_e + \cos\theta_e\,\boldsymbol{\Phi}_{x,e} + \sin\theta_e\,\boldsymbol{\Phi}_{y,e}. \tag{29}$$

The baseline limit is $R_{\min} = 20$ mm. The constraint is necessary for laying the field with tows but not sufficient: tow spreading and the resulting gaps or overlaps are treated separately (Sec. 5.7).

4.2 The design problems

Steered plate. For a target frequency $\omega_\star$,

$$\min_{\hat{\boldsymbol{\theta}}} \ \frac{J(\hat{\boldsymbol{\theta}})}{J_{\mathrm{UD}}} \quad \text{s.t.} \quad \frac{\omega_1(\hat{\boldsymbol{\theta}})}{\omega_\star} - 1 = 0, \qquad c_e(\hat{\boldsymbol{\theta}}) \ge 0 \ \ \forall e, \tag{30}$$

where J and ω_1 are those of the adapted-mode, relaxed model (Sec. 2.4). The frequency is an *equality*, so a gain can never come from a stiffer or softer panel. The mass is fixed by construction. We solve (30) by sequential quadratic programming (SLSQP [51]) from six starts (uniform $0^\circ, 20^\circ, 45^\circ, 70^\circ$ and two random fields), and keep the best result that satisfies both constraints to 10^{-4} (the residuals actually reached are 10^{-10} or smaller, Table 2).

Uniform stacks. We enumerate every symmetric eight-ply laminate $[\theta_4/\theta_3/\theta_2/\theta_1]_s$ with $\theta_i \in \{-75°, -60°, \dots, 90°\}$: $12^4 = 20{,}736$ stacks, each evaluated exactly on the same model. A uniform laminate needs no optimizer, because its operators are linear combinations of precomputed moment tensors (e.g. $\boldsymbol{K}_b = \sum_{ab} D_{ab} \boldsymbol{I}^K_{ab}$).

Steered stack. The best uniform stack at $\omega_\star$ is rotated pointwise, $\boldsymbol{C}(\boldsymbol{x}) = \boldsymbol{T}(\theta(\boldsymbol{x}))\boldsymbol{C}_0\boldsymbol{T}(\theta(\boldsymbol{x}))^\top$ for $\boldsymbol{C} \in \{\boldsymbol{A}, \boldsymbol{D}\}$, and (30) is solved for $\theta(\boldsymbol{x})$.

Beam. $\theta(x)$ is mirror symmetric about mid-span and piecewise linear on 13 knots over the half span. The tow limit is the slope bound $|\theta'| \le \kappa_{\text{tow}}$, and the frequency is held as $\omega_1 \ge f_{\text{hold}}\, \omega_{1,\text{UD}}$. A uniform-angle beam cannot change J (Corollary 3) and every steered beam is *softer* than UD, so an equality at $f_{\text{hold}} = 1$ admits only UD. We report $f_{\text{hold}} \in \{0.98, 0.90\}$ and $\kappa_{\text{tow}} \in \{900, 1800, \infty\}$ deg/m. The optimizer works on the condensed beam evaluator (adapted mode, exact axial condensation), and every design is *confirmed* on the full FE beam by its cubic SSM.

4.3 Sensitivities on the adapted mode

Let a perturbation $\delta\theta(\boldsymbol{x})$ change the stiffnesses by $\delta\boldsymbol{D} = \boldsymbol{D}'\delta\theta$ and $\delta\boldsymbol{A} = \boldsymbol{A}'\delta\theta$, with $\boldsymbol{C}' = \boldsymbol{T}'\boldsymbol{C}_0\boldsymbol{T}^\top + \boldsymbol{T}\boldsymbol{C}_0\boldsymbol{T}'^\top$. Let $(\lambda_j, \boldsymbol{v}_j)$ be the Ritz pairs of the bending block ($\lambda_1 = \omega_1^2$), with curvature fields $\boldsymbol{\kappa}_j$. The mass does not depend on θ.

Frequency. The Rayleigh quotient is stationary (Fox–Kapoor), so

$$\delta\omega_1^2 = \int \boldsymbol{\kappa}_1^\top \delta\boldsymbol{D}\, \boldsymbol{\kappa}_1. \tag{31}$$

Mode shape. The eigenvector derivative is

$$\delta\boldsymbol{v}_1 = \sum_{j\ge 2} \frac{\boldsymbol{v}_j^\top \delta\boldsymbol{K}_b \boldsymbol{v}_1}{\lambda_1 - \lambda_j}\boldsymbol{v}_j, \qquad \boldsymbol{v}_j^\top \delta\boldsymbol{K}_b\boldsymbol{v}_1 = \int \boldsymbol{\kappa}_j^\top \delta\boldsymbol{D}\, \boldsymbol{\kappa}_1. \tag{32}$$

Hardening. By Proposition 1, γ is a minimum over the in-plane state, so its total derivative needs *no in-plane adjoint*: only the explicit $\boldsymbol{A}$-dependence and the dependence on the mode (through $\boldsymbol{m}$) remain. With $\partial\gamma/\partial\boldsymbol{m} = \boldsymbol{A}\boldsymbol{\varepsilon}_r$ and $\partial\boldsymbol{m}/\partial v_k = [2w_{1,x}\phi_{k,x},\, 2w_{1,y}\phi_{k,y},\, 2(w_{1,x}\phi_{k,y} + w_{1,y}\phi_{k,x})]$,

$$\delta\gamma = \int \Big[\tfrac{1}{2}\boldsymbol{\varepsilon}_r^\top \delta\boldsymbol{A}\, \boldsymbol{\varepsilon}_r + \boldsymbol{\kappa}_\beta^\top \delta\boldsymbol{D}\, \boldsymbol{\kappa}_1\Big], \qquad \boldsymbol{\kappa}_\beta = \sum_{j\ge 2} \frac{\boldsymbol{v}_j^\top \partial_{\boldsymbol{v}}\gamma}{\lambda_1 - \lambda_j}\boldsymbol{\kappa}_j, \quad \partial_{v_k}\gamma = \int (\boldsymbol{A}\boldsymbol{\varepsilon}_r)^\top \partial_{v_k}\boldsymbol{m}. \tag{33}$$

The *adjoint curvature field* $\boldsymbol{\kappa}_\beta$ collapses every eigenvector term into one field.

Mid-plate value. In the same way, $\delta w_c = \int \boldsymbol{\kappa}_{\text{mid}}^\top \delta\boldsymbol{D}\, \boldsymbol{\kappa}_1$, with $\boldsymbol{\kappa}_{\text{mid}} = \sum_{j\ge 2} \frac{w_c(\boldsymbol{v}_j)}{\lambda_1 - \lambda_j}\boldsymbol{\kappa}_j$.

Detuning. Since $J = \gamma/(\omega_1^2 w_c^2)$,

$$\delta J = \frac{\delta\gamma}{\omega_1^2 w_c^2} - \frac{\gamma\, \delta\omega_1^2}{\omega_1^4 w_c^2} - \frac{2\gamma\, \delta w_c}{\omega_1^2 w_c^3}, \qquad \frac{\partial J}{\partial \hat{\boldsymbol{\theta}}} = \boldsymbol{\Phi}^\top \Big[w_q \frac{\partial J}{\partial \theta_q}\Big]_q, \tag{34}$$

where w_q are the quadrature weights. The whole gradient costs one eigensolve of dimension N_w, one in-plane solve of dimension $2N_s$, and pointwise contractions. It does not depend on M, so the support grid can be refined freely. Equation (34) is the local optimality condition of Proposition 5, plus the mode-shape corrections carried by $\boldsymbol{\kappa}_\beta$ and $\boldsymbol{\kappa}_{\text{mid}}$.

Table 2: Multi-start diagnostics of the held-frequency plate design (all SLSQP runs terminated normally; residuals at termination; generated by `r1_verification.py`). The reported design is the best start; 4 of six starts reach within 4.2% of it, and two stall in poorer local minima.

start	J/J_{UD}	$\omega_1/\omega_\star - 1$	$\min_e c_e$	iterations
uniform 0	0.6508	3×10^{-13}	-2×10^{-14}	22
uniform 20	0.3659	3×10^{-11}	8×10^{-12}	54
uniform 45	0.3813	9×10^{-14}	-4×10^{-15}	54
uniform 70	0.5711	3×10^{-13}	7×10^{-14}	44
random 1	0.3675	3×10^{-11}	1×10^{-12}	72
random 2	0.3675	2×10^{-11}	4×10^{-12}	79

4.4 Verification of the evaluator and its gradient

- **Gradient.** At the published design, central finite differences (steps 10^{-4}–10^{-6}) along 10 directions (five random, five largest-gradient coordinates) agree with the analytic derivatives of J, ω_1, γ and the tow curvature to better than 10^{-8} relative at the best of the three steps (and to better than 10^{-6} at every step) (`r1_verification.py`).
- **Constraints on an independent grid.** The tow curvature of the final field, evaluated analytically on a 1 mm grid that shares no point with the optimizer's quadrature, gives a minimum radius of 19.9 mm (limit 20 mm). The frequency equality is re-checked on finer bases and on the FE model in Sec. 5.5.
- **Multi-start.** Table 2 lists every start of the held-frequency solve.
- **Isotropy.** With $\boldsymbol{Q}$ isotropic, $\partial J/\partial\theta$ vanishes to 10^{-10} relative (Corollary 2).
- **Proposition 1.** $\int e_G = \gamma$ and $\int e_K = \omega_1^2$ hold to 10^{-10}.
- **Rotation.** The rotation (1) reproduces the classical $\bar{\boldsymbol{Q}}$ to machine precision, and Proposition 4 holds to 10^{-9} for random strains.
- **Uniform path.** The fast uniform-laminate path equals the general one to 10^{-13}.
- **Reference values.** For the UD plate the evaluator gives $f_1 = 256.3$ Hz and a 10.95% shift at $a/h = 0.4$, in agreement with the 10.96% of the assumed-modes model of the finite-element comparison (Appendix C).

Basis convergence is in Table 1.

The unit checks above are in `tests/test_relaxed_plate.py`; the finite-difference, independent-grid and multi-start checks are in `examples/r1_verification.py`.

5 Results

5.1 The one-dimensional test (P1, P2)

Uniform invariance. Rotating every ply of the symmetric beam uniformly to 20° or 45° lowers the linear frequency to 0.895 and 0.561 of the UD value (207.2 Hz). On the full FE model, the cubic SSM detuning coefficient is unchanged to four digits (ratio 1.0000), as Corollary 3 requires. No uniform single-ply angle can therefore change J at all.

Table 3: Symmetric steered beam: detuning ratio J/J_{UD} from the condensed evaluator (optimizer) and from the full FE + cubic SSM (confirmation), with the realized frequency ratio.

tow limit	frequency floor	J/J_{UD} condensed	J/J_{UD} FE+SSM	$\omega_1/\omega_{1,\mathrm{UD}}$ (SSM)
900°/m	0.98	0.818	0.813	0.979
900°/m	0.90	0.627	0.621	0.898
1800°/m	0.98	0.667	0.656	0.978
1800°/m	0.90	0.430	0.440	0.896
none	0.98	0.414	0.405	0.976
none	0.90	0.370	0.375	0.916

Steering. Table 3 and Fig. 2 report the optimized symmetric beams. Every design has $\boldsymbol{B} \equiv \mathbf{0}$, so the bending–extension coupling plays no role. The optimizer does exactly what Proposition 6 predicts: it turns the fibers off-axis in two zones centered, to within the knot spacing of the angle field ($x/L = 0.25,\ 0.75$), on the inflection points ($x/L = 0.224, 0.776$), where φ''^2 vanishes, and keeps them axial where the beam bends (Fig. 2b). The SSM of the full FE beam confirms the condensed evaluator's prediction for every design to within 0.011 in J/J_{UD}.

- **Tow-curvature limits.** At the 900°/m limit and $\omega_1 \geq 0.98\,\omega_{1,\mathrm{UD}}$, the detuning falls by 19%. At 1800°/m it falls by 34%, and with no tow limit by 59%.
- **Frequency allowance.** Relaxing the frequency to $0.90\,\omega_{1,\mathrm{UD}}$ gives 38%, 56% and 62% respectively.
- **The coupling alternative.** The intuitive alternative is an unsymmetric laminate whose bending–extension coupling $\boldsymbol{B}(x)$ injects a softening quadratic force. We optimize it with its own, related objective (the RMS backbone drift over $a/h \leq 0.4$; two independently steered plies with two sine modes each, the same tow limit, and $\omega_1 \geq 0.90\,\omega_{1,\mathrm{UD}}$). It reduces that drift by only 2%, at $0.90\,\omega_{1,\mathrm{UD}}$. The comparison is therefore not like-for-like in objective, but the gap (a few percent against 38% for symmetric steering at the same 900°/m limit and frequency floor) is far larger than the difference between the two objectives.

The lever on a clamped beam is therefore the spatial separation of curvature and stretch, *not* the bending–extension coupling. Its size is set by how narrow the compliant zones can be (Corollary 4). With no tow limit the two zones are each $\approx 0.05L$ wide at near-transverse fibers (total compliant length $\ell \approx 0.1L$), and (26) gives $J/J_{\mathrm{UD}} \approx 0.4$–$0.5$, against the computed 0.405. At 900°/m a 90° turn needs a third of the span, so the zones cannot both be narrow and reach low modulus. This is why that limit binds.

5.2 The achievable frequency–hardening set of the plate (P3)

Figure 3a places every design class in the plane ($\omega_1/\omega_{1,\mathrm{UD}}$, J/J_{UD}) at the same mass.

Uniform layups. A uniform single angle traces a one-parameter curve. As θ grows from 0° the plate stiffens (up to $1.52\,\omega_{1,\mathrm{UD}}$ at 90°, fibers across the short span), and J first falls to 0.57 at $\theta = 37°$, where $\omega_1 = 1.28\,\omega_{1,\mathrm{UD}}$, then rises. The 20,736 uniform symmetric stacks on a 15° grid fill a region with a global minimum of 0.50 at $1.55\,\omega_{1,\mathrm{UD}}$. A grid envelope is not the strongest uniform competitor: between grid points the continuous single angle already beats it (0.75 vs. 0.81 at $1.1\,\omega_{1,\mathrm{UD}}$). We therefore also optimize the four ply angles of $[\theta_4/\theta_3/\theta_2/\theta_1]_s$ *continuously* at each held frequency (SLSQP with the frequency equality, 22 starts from the best grid stacks,

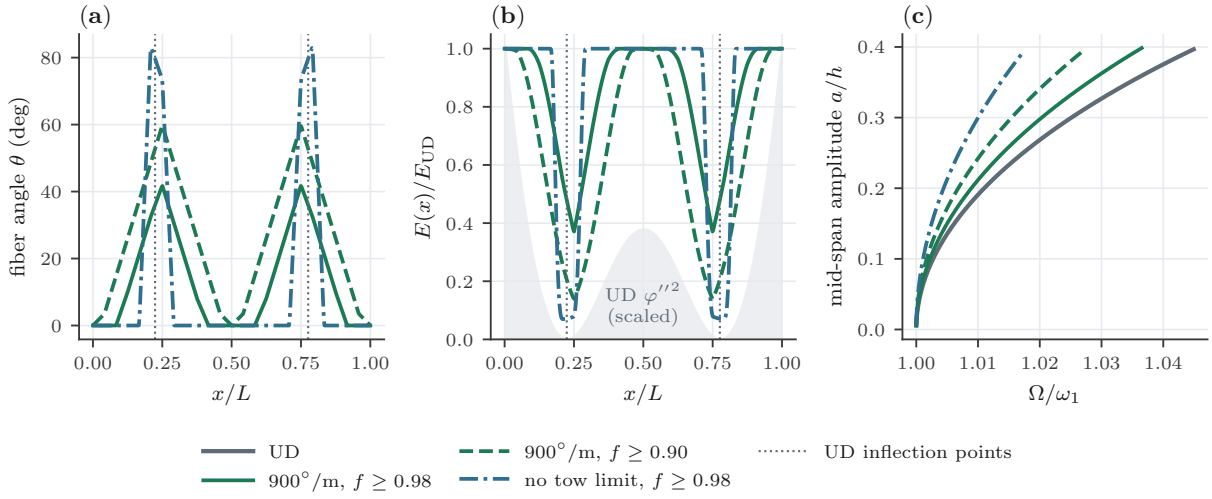


Figure 2: **On a symmetric beam, steering flattens the backbone by turning the fibers compliant where the beam does not bend. (a)** Optimized mirror-symmetric fiber angle of the symmetric ($\boldsymbol{B} = \mathbf{0}$) clamped beam for three designs: tow limit 900°/m with the frequency held to $f \geq 0.98$ or $f \geq 0.90$ of the UD value, and no tow limit with $f \geq 0.98$. The UD beam ($\theta \equiv 0$, $E = E_{\mathrm{UD}}$) appears in (c) only. **(b)** The axial modulus these fields produce, against the UD curvature weight φ''^2 that sets ω_1^2 (shaded). Fibers turn compliant where the beam does not bend: the inflection points (dotted). **(c)** Mid-span backbones from the full FE beam and its cubic SSM; steering flattens the backbone markedly. (The optimized unsymmetric, $\boldsymbol{B}$-coupled laminate was computed on a different beam model and objective, so it is compared in the text, not plotted.)

the single-angle solution and random stacks; `stack_continuous.py`). The best uniform laminate reaches $J/J_{\mathrm{UD}} = 0.74$, 0.61 and 0.55 at 1.1, 1.2 and $1.3\,\omega_{1,\mathrm{UD}}$.

The UD floor. Corollary 1 proves that the lowest-frequency uniform symmetric laminate of this plate is a single-angle one, and the single-angle sweep shows that it is UD. The enumeration agrees: the smallest stack frequency is $1.000\,\omega_{1,\mathrm{UD}}$. At exactly $\omega_1 = \omega_{1,\mathrm{UD}}$ the continuous stack optimum is UD itself (all ply angles $\to 0$); the best grid stack within ±1% of that frequency, at a slightly different ω_1, has $J/J_{\mathrm{UD}} = 0.997$, a negligible difference. For a plate mandated to the UD frequency, the uniform design space is therefore a single point. This is the one comparison that is exact for *every* uniform laminate. Above $\omega_{1,\mathrm{UD}}$ the uniform competitors are the classes we surveyed: all single angles, all 20,736 grid stacks and the continuously optimized eight-ply stacks. Those comparisons are numerical, not proved bounds.

Steered designs. At exactly $\omega_{1,\mathrm{UD}}$, the steered single ply reaches $J/J_{\mathrm{UD}} = 0.366$, a 63% reduction, with $R_{\min} = 20\,\mathrm{mm}$ imposed (it is active but changes J only in the fourth digit: without it the optimum is 0.366, Sec. 5.4). The steered designs lie well below the best surveyed uniform laminate at every frequency we test: 0.29, 0.24 and 0.21 at 1.1, 1.2 and $1.3\,\omega_{1,\mathrm{UD}}$, against 0.74, 0.61 and 0.55 for the best continuous stacks. Steering can also *lower* the frequency below the uniform floor (0.55 at $0.9\,\omega_{1,\mathrm{UD}}$), which, by Corollary 1, no uniform laminate can do. Steering the best stack at $\omega_{1,\mathrm{UD}}$ gives 0.37, no better than the steered single ply. For this plate and objective, the in-plane channel of Sec. 3.3 dominates the through-thickness channel. Prediction P3 holds with a wide margin: exactly at $\omega_{1,\mathrm{UD}}$, and against every surveyed uniform class above it.

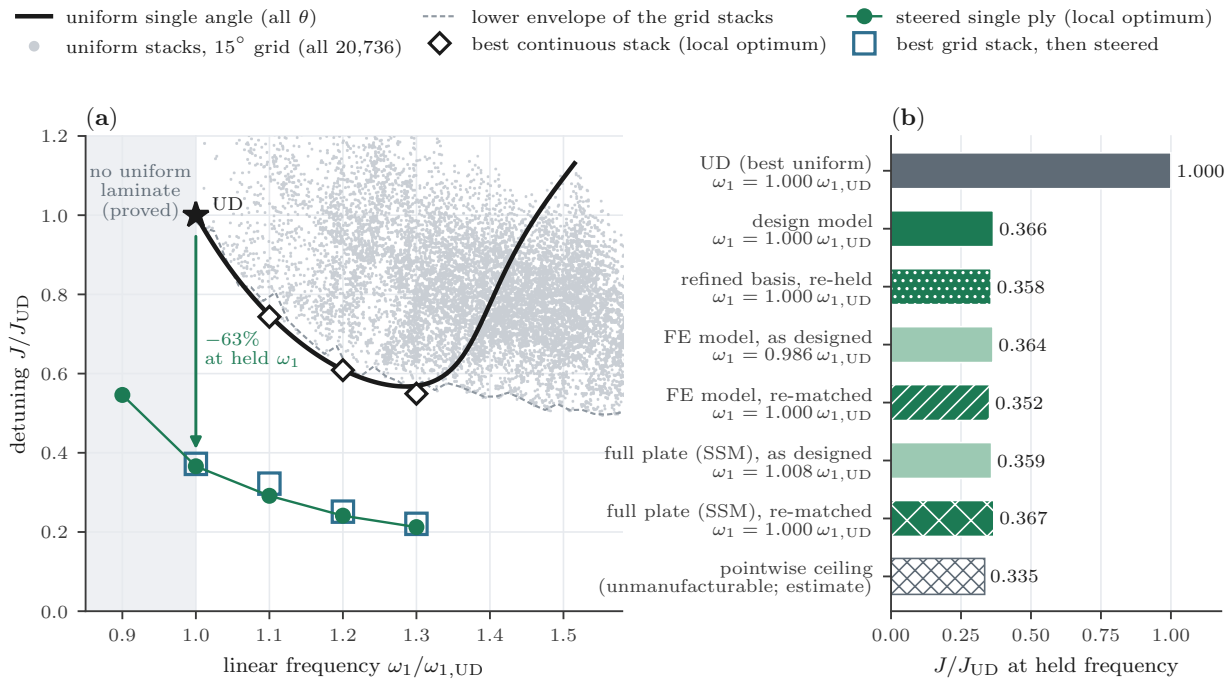


Figure 3: **At the unidirectional frequency and mass, steering reaches a hardening that no uniform symmetric laminate can reach, and above it steering undercuts every uniform class surveyed; the held-frequency result holds with the frequency re-matched in every model. (a)** Linear frequency vs. frequency-normalized hardening. Provenance is marked in the legend. *Proved*: the shaded band $\omega_1 < \omega_{1,\mathrm{UD}}$, where no uniform laminate exists (Corollary 1). *Exhaustive*: the uniform single angle (black curve) and all 20,736 symmetric stacks on a 15° grid (gray dots, clipped at $J/J_{\mathrm{UD}} = 1.2$; dashed: their lower envelope). *Local optima*: the best continuous eight-ply stack at each held frequency (open diamonds, 22 starts), the steered single ply at five target frequencies with $R_{\min} = 20\,\mathrm{mm}$ (green), and the best grid stack at each target, then steered (blue squares; none exists at $0.9\,\omega_{1,\mathrm{UD}}$). The arrow marks the held-frequency reduction. **(b)** The held-frequency design's J/J_{UD} in each model, with the frequency at which it is evaluated: the design model (equality exact); a refined basis ($m, n \le 12$) with the equality re-imposed; the independent FE model as designed (ω_1 1.4% low) and with the frequency re-matched; the full multi-mode plate's SSM as designed (ω_1 0.8% high) and re-matched; and the pointwise ceiling of Sec. 5.4 (unmanufacturable; an estimate). At $\omega_{1,\mathrm{UD}}$ the best uniform laminate is UD itself; the best grid stack there is UD to within 0.3% in J ($[0/0/0/15]_s$), and steering it gives 0.37.

5.3 The mechanism (P4)

Figure 4 tests the mechanism of Sec. 3 directly.

- **Where the energies live.** Both energies of the UD mode peak at the center. The bending energy is concentrated there, with the principal curvature along y, across the short span, so the held-frequency constraint favors fibers along x there (panel a). The relaxed-stretch energy lies in a band along x through the center, with its across-stretch direction along x there and turning toward the corners (panel b).
- **Sign of the trade-off.** At the held UD frequency the multiplier is negative, $\Lambda = -0.16$

(dimensionless: e_G and e_K normalized by the UD γ and ω_1^2): minimizing the stretch energy alone would raise the frequency to $1.20\,\omega_{1,\mathrm{UD}}$, so the constraint pushes the fibers *across* the curvature. At the center the two preferences agree (fibers along x), and the pointwise design $\theta^\star$ keeps the UD direction there.

- **Conflict.** Away from the center the across-stretch and across-curvature directions separate (panel c); their energy-weighted mean angle is 33°. This is where a single angle must compromise.
- **Pointwise prediction.** The first-order design $\theta^\star(\boldsymbol{x})$ of Proposition 5, computed from the UD fields alone with no optimizer and no smoothness (panel d), already gives $J/J_{\mathrm{UD}} = 0.46$ when evaluated on the full model. Its frequency is $0.935\,\omega_{1,\mathrm{UD}}$, because the frozen-field model ignores the adaptation of the mode.
- **Optimized field.** The optimized, curvature-limited field (panel e) shares the large-scale pattern of $\theta^\star$ (fibers turned off-axis toward the short edges) but departs from it in detail: the energy-weighted RMS angle difference is 29° over the plate and 28° where the energies are large, and at the energy peak the optimum is tilted to −21°. The frozen-field design ignores the adaptation of the mode and the smoothness and tow limit of a curvature-feasible field, both of which the optimizer must respect.

Panel f shows which terms move, normalized by UD:

- the raw (restrained) membrane cubic falls to $k_{nl} = 0.52$;
- the relaxation $2\boldsymbol{g}^\top \boldsymbol{K}_{\mathrm{in}}^{-1}\boldsymbol{g}$ changes to 0.81 of its UD value, a share of k_{nl} rising from 33% to 51%;
- the relaxed cubic falls to $\gamma = 0.38$;
- the frequency is held ($\omega_1^2 = 1.00$) and the mode's center value changes to $w_c = 1.01$.

The hardening falls mainly through the direct term of (13): fibers across the principal stretch store less membrane energy for the same stretch, so k_{nl} falls to 0.52 of its UD value. The relaxation falls less (to 0.81), so its share of the stretch energy rises and γ falls further than k_{nl} alone. Prediction P4 holds partly: $\theta^\star$ alone captures most of the gain, though at $0.935\,\omega_{1,\mathrm{UD}}$, but the optimum departs from it by 28° RMS even where the energies are large.

5.4 What bounds the lever (P5)

Anisotropy. Along the family $\boldsymbol{Q}(s) = \boldsymbol{Q}_{\mathrm{iso}} + s(\boldsymbol{Q} - \boldsymbol{Q}_{\mathrm{iso}})$, which keeps the isotropic invariants U_1, U_4, U_5 and scales U_2, U_3 by s, the held-frequency steered optimum falls monotonically with the anisotropy:

- $J/J_{\mathrm{UD}} = 0.98$ at $E_1/E_2 = 1.2$;
- 0.84 at 2.9;
- 0.37 at 14.6 (CFRP);
- 0.21 at 31.9 (Fig. 5a).

The isotropic limit has no lever (Corollary 2).

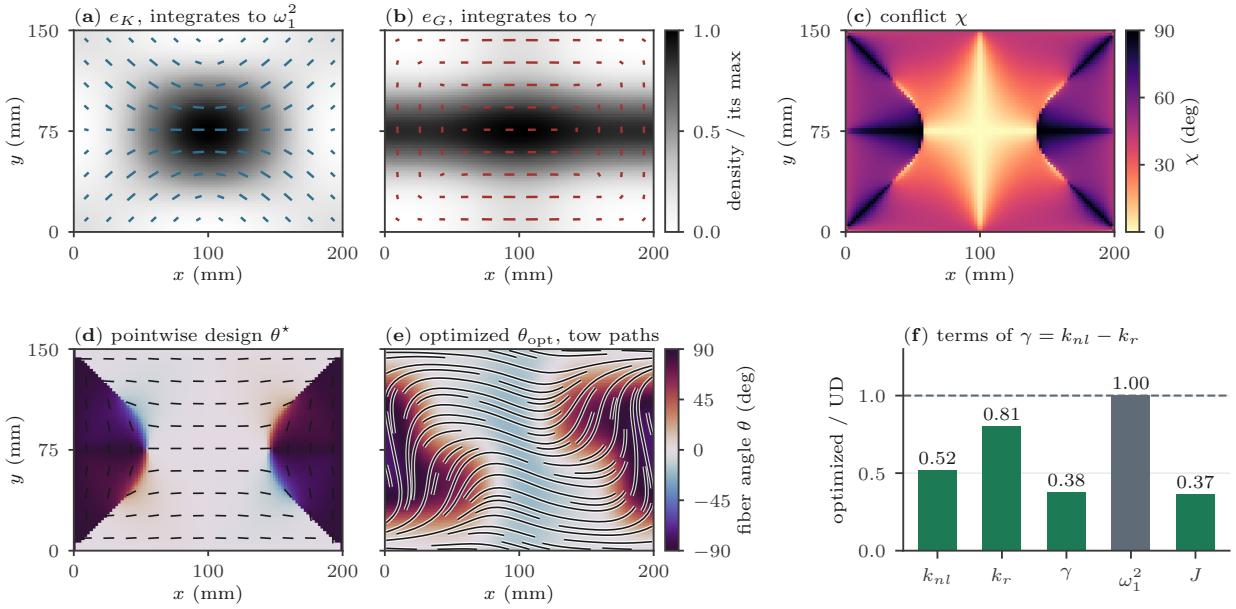


Figure 4: **The two energies prefer different fiber directions away from the plate center; the optimized field follows a compromise that a single angle cannot make.** All maps show the full 200 mm × 150 mm plate with the same axes; angles are measured from the x axis. **(a)** Bending-energy density e_K of the UD fundamental, which integrates to ω_1^2. Blue ticks: the fiber direction the held-frequency constraint favors ($\Lambda < 0$: across the principal curvature, $\varphi_K + 90°$). **(b)** Relaxed-stretch density e_G, which integrates to γ, on the same normalized gray scale. Red ticks: the direction that minimizes it (across the principal stretch, $\varphi_G + 90°$). Tick length in (a) and (b) scales with the local anisotropic amplitude. **(c)** Conflict χ, the angle between the ticks of (a) and (b). It vanishes at the center and along the central cross ($x = 100$ mm, and $y = 75$ mm for $60 < x < 140$ mm), where one direction serves both energies, and reaches 90° along $y = 75$ mm near the short edges. The sharp ridges are lines where one of the two principal directions is undefined (equal principal strains). **(d)** First-order pointwise design $\theta^\star$ from the UD fields alone (Proposition 5); black ticks: the same direction field. **(e)** Optimized curvature-limited field; colors give the fiber angle and black lines the tow paths. (d) and (e) share one cyclic color scale, lightest at $\theta = 0°$. **(f)** Optimized/UD ratios of the terms of (13): the restrained cubic k_{nl}, the relaxation $k_r = 2\boldsymbol{g}^\top \boldsymbol{K}_{\text{in}}^{-1}\boldsymbol{g}$, their difference γ, ω_1^2 (held) and J; dashed: parity with UD.

Tow curvature. The held-frequency optimum degrades smoothly as the minimum tow radius grows:

- 0.39 at $R_{\min} = 30$ mm;
- 0.47 at 50 mm;
- 0.57 at 80 mm;
- 0.71 at 150 mm;
- 0.84 at 400 mm (Fig. 5b).

Below $R_{\min} \approx 20$ mm it saturates at 0.37. Two checks show that the saturation is physical rather than an artifact of the 50 mm support grid.

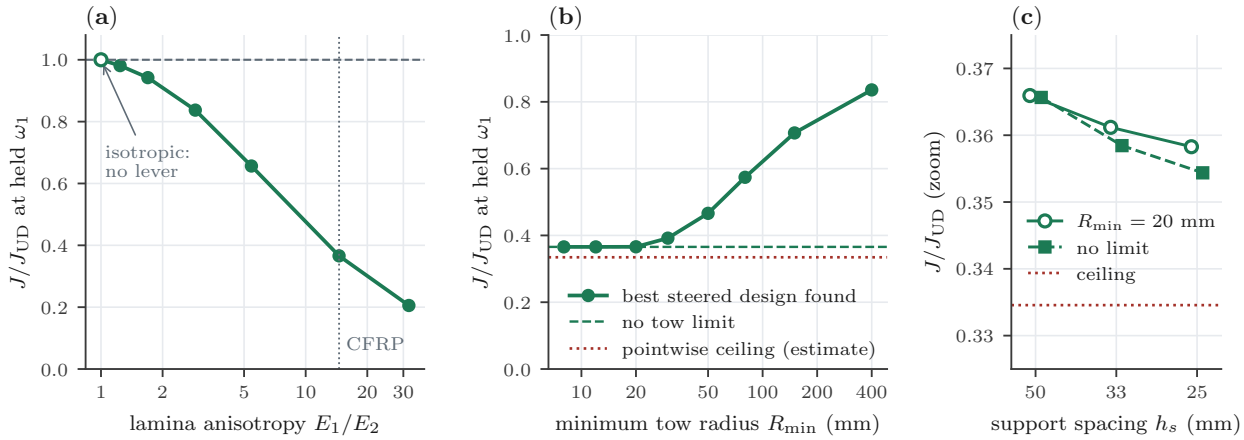


Figure 5: **The lever vanishes for isotropic material, grows with anisotropy, and saturates once tows can turn on the strain-field length scale.** All at held frequency. **(a)** Steered optimum vs. lamina anisotropy along a family that scales only the anisotropic invariants ($E_1/E_2 = 1$ is isotropic, where $J/J_{\mathrm{UD}} = 1$ exactly by Corollary 2, so that point is not computed; dashed: parity with UD; dotted: CFRP). **(b)** Tow-radius front: best design found vs. minimum tow radius on the 50 mm support grid; dashed: the same grid with no tow limit; dotted: the pointwise ceiling (unconstrained, unsmoothed; an estimate, not a bound). Each point is the best of six starts. **(c)** Support-grid refinement (zoomed ordinate; the three support spacings are plotted as equally spaced categories, with the two series offset slightly for visibility): at $R_{\mathrm{min}} = 20$ mm (open) and with no tow limit (filled) the optimum moves by less than 0.02 and approaches the ceiling, so the saturation in (b) is physical, not a parametrization artifact.

- **Finer grid.** Refining the grid to 33 mm and 25 mm changes the optimum only to 0.361 and 0.358 at $R_{\mathrm{min}} = 20$ mm, and to 0.358 and 0.354 with no tow limit.
- **Pointwise ceiling.** Freeing θ at every quadrature point, with no smoothness and no tow limit, gives 0.33. This field cannot be laid, and it is the best local optimum found for a nonconvex problem, so it estimates rather than bounds what single-ply steering can reach.

The plate's strain fields vary on a length scale of a few centimeters. Once tows can turn on that scale, further manufacturing freedom buys little. The front of Fig. 5b is the best design found from six starts at each radius, so the attainable front lies at or below it; the pointwise ceiling indicates how much lower it could plausibly go. Prediction P5 holds: the lever is set by the material anisotropy and by the ratio of the tow radius to the length scale of the mode's strain fields. Its third ingredient, the conflict between the two fields, is tested across plates in Sec. 5.7.

5.5 Independent confirmation on the full model

The design was produced on the condensed, adapted-mode evaluator. We confirm it on the full multi-mode plate: 36 transverse and 72 in-plane modes, in-plane inertia retained, an independent Gauss-quadrature assembly, and amplitudes measured as the true mid-plate deflection (Table 4).

- **Full SSM.** The full cubic SSM gives detuning coefficients β within 1.2% of the evaluator for both designs. The detuning ratio is 0.359, a 64% reduction.
- **EPMC.** Full-order EPMC continuation ($H = 3$ harmonics) gives a ratio of 0.362.

Table 4: Confirmation of the held-frequency design. β: small-amplitude coefficient of $\Omega/\omega_1 - 1 = \beta(a/h)^2$ at the plate center.

model	f_1 UD (Hz)	f_1 steered (Hz)	β UD	β steered
design evaluator (64+72 modes, condensed)	256.3	256.3	0.684	0.250
full plate, cubic SSM (36+72 modes)	256.3	258.3	0.689	0.248
full plate, EPMC ($H = 3$)	—	—	0.680	0.246
independent FE (ACM + Q4, 48×36, App. C)	256.2	252.6	0.685	0.250

- **Frequency.** The steered design's full-model frequency is 258.3 Hz against 256.3 Hz for UD. The difference is the Ritz upper-bound convergence of the transverse basis for the steered field (Table 1: the steered frequency falls by 0.9% from $m, n \leq 8$ to $m, n \leq 16$ while the UD frequency is unchanged), not a design change.
- **Discretization.** An independent finite-element model (ACM Kirchhoff bending with Q4 membrane elements, Appendix C) evaluated on the optimized field gives a detuning ratio of 0.364. Across the three discretizations the steered design's frequency lies within $\pm 1.4\%$ of the UD value.

Matched-frequency comparison. In the refined design basis and in the FE model the steered frequency comes out slightly *below* the UD value, which is unfavorable to steering because J falls as the frequency rises along the steered front (Fig. 3a). In the full multi-mode plate it comes out 0.8% *above*, which favors it. To remove the drift in both directions rather than bound it, we re-impose the equality in each model (`r1_matched_frequency.py`, `r1_full_matched.py`; Fig. 3b).

- **Refined basis.** On $m, n \leq 12$ (144 transverse and 128 in-plane modes) the published design has $\omega_1 = 0.993\,\omega_{1,\mathrm{UD}}$ and $J/J_{\mathrm{UD}} = 0.364$. Re-solving the held-frequency problem on that basis, with the equality exact, gives $J/J_{\mathrm{UD}} = 0.358$.
- **Independent FE model.** The design as laid has $\omega_1 = 0.986\,\omega_{1,\mathrm{UD}}$ and ratio 0.364. We move it along the design model's frequency-gradient direction until the FE frequencies agree to 10^{-5} (a one-parameter move, not a re-optimization, so conservative). The ratio is then 0.352. Along that path $\mathrm{d}(J/J_{\mathrm{UD}})/\mathrm{d}(\omega_1/\omega_{1,\mathrm{UD}}) = -0.86$, so a frequency drift of 1% moves the headline ratio by less than 0.01.
- **Full multi-mode plate.** Moved in the same way until its linear frequency equals the UD value to 10^{-6}, the steered design has an SSM detuning ratio of 0.367 (as designed: 0.359) and an EPMC ratio of 0.370.

The held-frequency result is therefore not an artifact of the design discretization. With the frequency matched in each model, the steered plate is at 0.352–0.370 of the UD hardening.

5.6 Consequence: a wider jump-free range

The backbone improvement (Secs. 5.2–5.5) is a property of the conservative system. Turning it into a forced-response statement needs a damping model and a forcing, so the absolute numbers below are specific to the model ($\zeta = 2\%$ mass-proportional, a center point load). The *ratio* of jump-free amplitudes is not: Proposition 3 predicts that the held-frequency reduction widens the jump-free amplitude by $(J_{\mathrm{steered}}/J_{\mathrm{UD}})^{-1/2} = 1.67$, whatever ζ is, and Sec. 5.7 tests that independence. We compute a_c numerically as the drive at which the full-order damped SSM forced response ($\zeta = 2\%$,

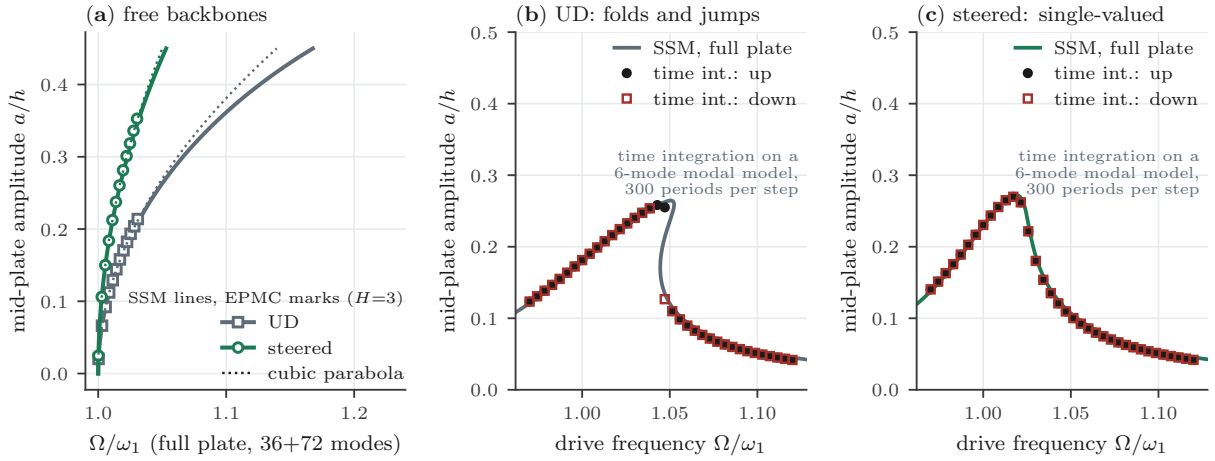


Figure 6: **The full model confirms the reduction, and at one drive the UD panel jumps while the steered panel, at the same frequency and mass, does not. (a)** Free mid-plate backbones of the full multi-mode plate (36 transverse + 72 in-plane modes): cubic SSM (lines) and full-order EPMC with $H = 3$ harmonics (marks; squares UD, circles steered). Dotted: the cubic parabola $1 + \beta(a/h)^2$ with the full-model SSM β (within 1.2% of the condensed evaluator's). **(b,c)** Forced response at one drive between the two jump thresholds ($\zeta = 2\%$, center point load): full-plate damped SSM (line) and direct time integration of up sweeps (black dots) and down sweeps (red squares) on a six-mode modal model, 300 forcing periods per step. A point is kept once its amplitude is steady to 0.4%, which excludes one UD point at the fold.

center point load) first becomes multivalued. It rises from $a_c/h = 0.208$ to 0.347, a factor 1.66 in amplitude and 1.64 in drive. This agrees with the predicted ratio, and the absolute thresholds agree too: (16) with the full-model β gives 0.211 and 0.353.

Direct time integration then shows the jump itself (Fig. 6b,c). We drive both panels with the same point load, set between their two thresholds, and sweep the frequency up and down. The UD panel is hysteretic: the up and down sweeps differ by up to $0.128\,h$ across the fold. The steered panel, at the same frequency and mass, is single-valued: its sweeps coincide to $0.000\,h$ and follow the SSM forced response. A finer sweep (41 steps over $\Omega/\omega_1 \in [1.00, 1.08]$; `confirm_fine_sweep.py`) confirms both: the UD up and down sweeps differ at 2 consecutive steps across the fold (by up to $0.125\,h$), the steered sweeps at none.

5.7 Robustness of the lever

The result above is for one plate, one support, one damping law and an idealized laminate. This section varies each in turn (Fig. 7).

Other plates and the conflict test of P5. We solve the same held-frequency problem ($R_{\min} = 20\,\text{mm}$, 50 mm support grid, six starts) for a square and a 2:1 plate, for clamped edges, and for in-plane traction-free edges, and compare each with the strongest uniform laminate at the same frequency (all single angles and a continuous eight-ply stack optimization; `r1_generality.py`).

- **Aspect ratio.** The steered optimum is $J/J_{\text{UD}} = 0.22$ on the square plate and 0.76 on the 2:1 plate, against 0.366 on the paper's 4:3 plate. On all three the UD plate is again the lowest-frequency uniform laminate, so UD is the best uniform design at $\omega_{1,\text{UD}}$.

- **Clamped edges.** Here UD is *not* the frequency floor: the lowest uniform frequency, $0.981\,\omega_{1,\mathrm{UD}}$, occurs at a single angle of 22°, as Corollary 1 allows. A real uniform competitor therefore exists at $\omega_{1,\mathrm{UD}}$. It reaches 0.89, and the steered field reaches 0.40.
- **In-plane movable edges.** With traction-free in-plane edges the UD plate hardens 14 times less, because 95% of its stretch energy relaxes. Steering still halves what remains (0.49), raising the relaxed share to 99%.

The lever is general, but its size varies by a factor of three, and the mechanism says why. Proposition 5 predicts that steering gains least where the two strain fields share their principal directions. The energy-weighted conflict χ of each UD plate (`r1_conflict.py`) orders the results: 21° on the 2:1 plate, whose fundamental bends nearly cylindrically across the short span (0.76); 33° on the 4:3 plate (0.366); and 48° on the square plate (0.22). The clamped (27°) and movable-edge (27°) plates fall near the same trend (Fig. 7b). The test is suggestive rather than decisive. The conflict was added to P5 in this revision, only three simply supported plates order it, and another property of the UD fields, the frequency that minimizing the stretch energy alone would select (1.23, 1.20 and $1.09\,\omega_{1,\mathrm{UD}}$ for the three plates), orders them equally well. The 2:1 value is also limited by the support grid, which has only three rows across its 100 mm span: refining the grid to 33 mm and 25 mm lowers it to 0.73 and 0.71, against 2% on the 4:3 plate. The ordering of the plates is unchanged.

Damping level and law. We recompute the jump-free amplitude of both designs on the full damped SSM (`r1_damping.py`) for mass-proportional damping $\zeta = 0.5$, 1, 2 and 4%, and for stiffness-proportional damping at 2%. The absolute thresholds scale as $\zeta^{0.49}$ (UD: $a_c/h = 0.106$ to 0.290), as Proposition 3's $\zeta^{1/2}$ predicts. The steered/UD ratio is 1.663–1.665 for every mass-proportional level and 1.69 for stiffness-proportional damping (Fig. 7c). Two caveats apply. With mass-proportional damping the cubic damped SSM is the model for which Proposition 3 is exact, so the constancy of the ratio there is largely a consistency check; the stiffness-proportional case, whose decay rate depends on amplitude through the mode, is the independent test. And at $\zeta = 4\%$ the thresholds ($a_c/h = 0.290$ for UD, 0.482 steered) exceed the amplitude range over which the cubic SSM is verified against EPMC ($a/h \le 0.21$ and ≤ 0.35), so those two points are cubic-model predictions only. Within those limits, the widening of the jump-free range is a property of the hardening, not of the damping model.

As built. The designs are idealized tow fields (Sec. 1). Laying them with tows of constant width along the tow paths changes the local tow density, because neighboring paths spread or converge at the rate $\mathrm{d}\ln(\text{spacing})/\mathrm{d}\ell = \nabla\cdot\boldsymbol{n}$, with $\boldsymbol{n} = (\cos\theta, \sin\theta)$. A course that starts fully covering its inflow edge reaches $\boldsymbol{x}$ with relative density $\tau(\boldsymbol{x}) = \exp(-\int \nabla\cdot\boldsymbol{n}\,\mathrm{d}\ell)$, integrated back along the path (`r1_asbuilt.py`). For the held-frequency design τ spans 0.03 to 22 (Fig. 7d). Uncut courses are therefore not an option: they would leave 85% of the plate thinned. In a constant-tow-count model (thickness $h\tau$, as in continuous tow shearing [37]) the frequency would fall to $0.42\,\omega_{1,\mathrm{UD}}$ at 0.43 of the mass, and J/J_{UD} would rise to about 100. The field requires tow cut/add. With cut/add inside courses of N tows, the coverage mismatch at a course boundary never exceeds one tow width and averages half of one. We model this, in the spirit of the defect-layer method [36], as a defect area fraction $f_d = \frac{1}{2N}\min\bigl(1, |\ln\tau|/\ln(1+1/N)\bigr)$. It is either resin-filled gaps (0% coverage: constant thickness, fiber fraction $1 - f_d$) or overlaps (100% coverage: thickness $1 + f_d$). Fig. 7e places the results in the frequency–hardening plane.

- With $N = 8$ tows per course, gaps give $J/J_{\mathrm{UD}} = 0.371$ at $0.982\,\omega_{1,\mathrm{UD}}$ (mass -1.5%), and overlaps give 0.324 at $1.062\,\omega_{1,\mathrm{UD}}$ (mass $+6.1\%$).
- Over $N = 4$–16, J/J_{UD} stays between 0.29 and 0.38, but the frequency and mass are no longer held: the frequency ranges over 0.96–$1.12\,\omega_{1,\mathrm{UD}}$ and the mass over 0.97–1.12 of UD. The lowest ratio, 0.29, belongs to the heaviest and stiffest variant (overlaps, $N = 4$). The as-built points lie near the steered front of the idealized designs, which is computed at fixed mass, so the comparison with that front is indicative only.
- A smooth random angle error of 2° RMS (correlation length 25 mm, 40 samples) gives $J/J_{\mathrm{UD}} = 0.365 \pm 0.007$, with a frequency scatter of 0.7%.

Defects mainly shift the frequency and mass and leave J/J_{UD} near the steered front; the defect model, in particular its cap $f_d \le 1/(2N)$, is an assumption that largely sets this result. A buildable design at an exactly held frequency would put the defect model inside the optimization, which is beyond the present scope (Sec. 6).

The optimizer. The analytic gradients of J, ω_1, γ and the tow curvature agree with central finite differences to better than 10^{-8} relative along 10 directions. The final field's tow curvature, re-evaluated on an independent 1 mm grid, gives a minimum radius of 19.9 mm against the 20 mm limit (an excess of 0.4% between quadrature points). Of the six starts, 4 reach within 4.2% of the best design and two stall at up to 0.65 (Table 2). The local optimizer can stall, and the reported optimum is the best of the starts (`r1_verification.py`).

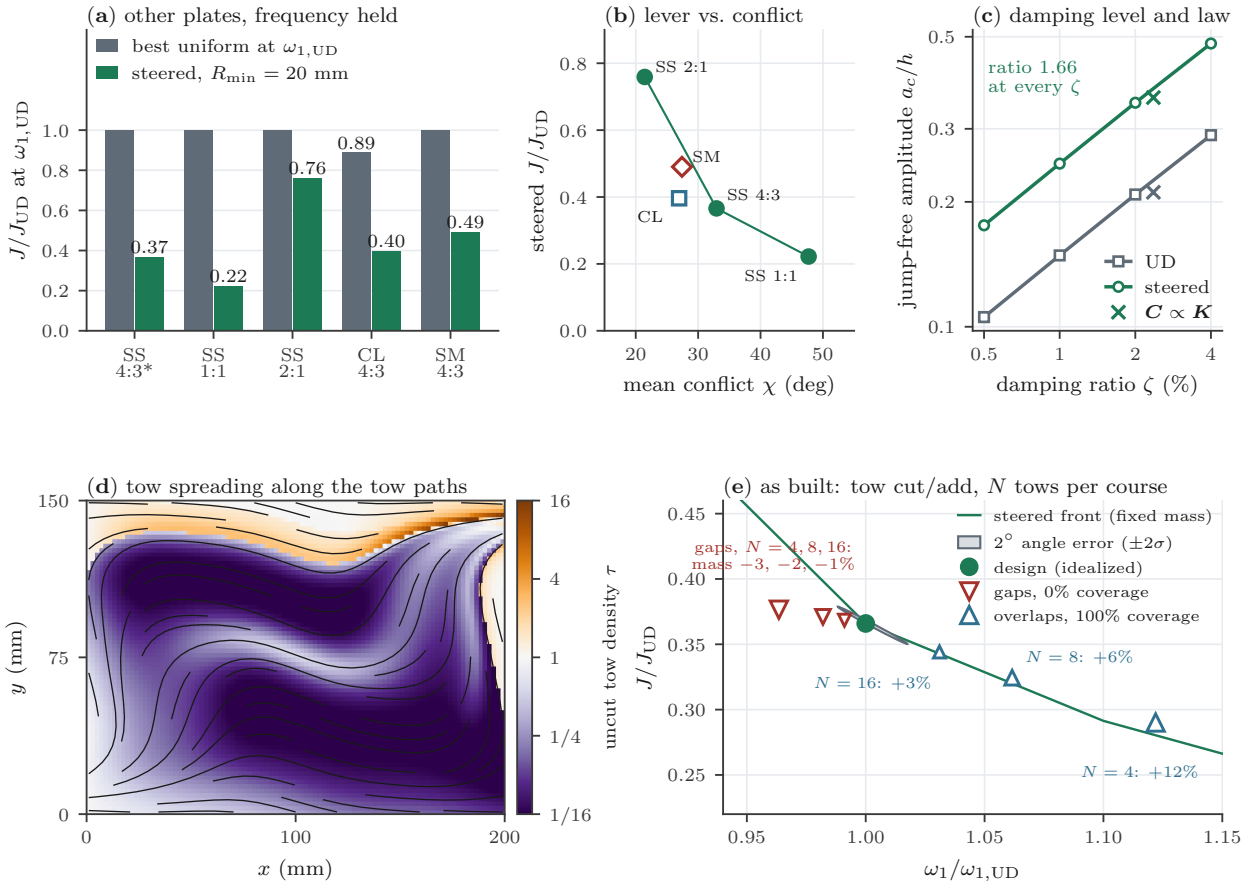


Figure 7: **The lever persists across plates, supports, damping laws and as-built defects, and its size is ordered by the conflict between the two strain fields. (a)** Held-frequency J/J_{UD} of the steered optimum (green) and of the best uniform laminate at the same frequency (gray) for simply supported (SS) plates of aspect ratio 4:3 (the paper's plate, $*$), 1:1 and 2:1, a clamped plate (CL) and a simply supported plate with in-plane movable edges (SM). Each plate is normalized by its own UD design; only the clamped plate has a uniform competitor other than UD at $\omega_{1,\mathrm{UD}}$. **(b)** Steered optimum vs. the energy-weighted conflict χ of the UD fields; line: the three SS plates. **(c)** Jump-free amplitude a_c/h from the full damped SSM vs. damping ratio (mass-proportional, lines; stiffness-proportional at $\zeta = 2\%$, crosses, drawn slightly to the right for visibility). Both designs scale as $\zeta^{1/2}$, and their ratio does not change. **(d)** Relative density τ of uncut constant-width tows laid along the tow paths of the held-frequency design (black lines). Purple: spreading (gaps if uncut); orange: convergence (overlaps); color scale clipped at 1/16 and 16 (τ spans 0.03–22). **(e)** As-built laminates with tow cut/add, $N = 4$, 8, 16 tows per course (labels: N and the mass change): resin gaps (down triangles), overlaps (up triangles), and a 2° RMS angle error (gray ellipse: mean ± 2 standard deviations). Their mass differs from UD, unlike the steered front, which is at fixed mass.

Table 5: Status of the predictions.

prediction	evidence	verdict
P1 isotropic material has no lever	Cor. 2; gradient $\equiv 0$ to 10^{-10}; Fig. 5a	holds (exact)
P2 uniform angle cannot change a single-ply beam; steering can, at the inflection points	SSM ratio 1.0000; Tab. 3; Fig. 2	holds
P3 steered plate reaches (ω_1, J) no uniform layup reaches	Fig. 3: exact at $\omega_{1,\mathrm{UD}}$ (Cor. 1); above it against all surveyed classes; persists with the frequency re-matched	holds
P4 optimized field follows preferred directions; $\theta^\star$ captures most of the gain	Fig. 4; $\theta^\star$: 0.46 at $0.935\,\omega_{1,\mathrm{UD}}$; optimum departs by 28° RMS where energies are large	holds partly
P5 lever grows with anisotropy and conflict, saturates at the field length scale	Fig. 5; Fig. 7a,b (five plates, ordered by χ)	holds
consequence: a_c widens as $J^{-1/2}$	1.66 vs. 1.67; time-domain sweeps; independent of ζ and damping law (Fig. 7c)	holds

6 Discussion

6.1 What is proved, what is computed, and what is not established

Table 5 summarizes the evidence for each prediction.

- **Proved** (for the stated model classes). Orientation enters each energy density only through U_2, U_3 and the local principal directions (Proposition 4). Isotropic material has no lever (Corollary 2). A single-ply beam's J is invariant under any uniform angle (Corollary 3). The lowest-frequency uniform symmetric laminate is a single-angle one (Corollary 1). A uniform angle that violates (24) is not stationary among fields in the frozen-field model (Proposition 5), and the jump-free amplitude scales as $J^{-1/2}$ at the cubic order (Proposition 3).
- **Computed**, with the stated discretizations and optimizer. The size of every gain: $J/J_{\mathrm{UD}} = 0.366$ at held frequency (between 0.352 and 0.370 with the frequency re-matched in refined and independent models), the margin over the surveyed uniform classes above $\omega_{1,\mathrm{UD}}$, the dose–response in anisotropy and tow radius, the generality across plates and its ordering by the conflict χ, and the as-built sensitivity. These are best-found values of a local optimizer (Table 2). The steered values are therefore upper bounds on the attainable hardening, and the uniform values above $\omega_{1,\mathrm{UD}}$ are upper bounds on the best uniform laminate. For mass-proportional damping the damping tests are consistency checks of the cubic SSM, and the as-built results rest on an assumed defect model.
- **Not established.** That the reported fields are global optima. That no uniform laminate outside the surveyed classes (more plies, a lamination-parameter search) does better above $\omega_{1,\mathrm{UD}}$; the steered values are at most 40% of the present uniform optimum. Performance of a physical panel.

6.2 Relation to prior work

The tools are not new: SSM reduction and adjoint SSM sensitivities [3, 5, 14, 15, 18], lamination parameters [21], and curvature-constrained orientation fields [50]. What is new is their target and the explanation. Fiber paths have been optimized for the linear frequency, and the nonlinear

response of the optimum was then examined [31, 32]. The nonlinear free, forced and thermal vibrations of prescribed curvilinear-fiber plates have been computed [45, 27, 28, 29, 30, 33]. Here the hardening is the objective, the linear frequency an equality constraint, and the result a mechanism with falsifiable predictions. On the clamped beam, the operative lever is not the bending–extension coupling but the harmonic-versus-curvature-weighted mean of Proposition 6, which needs no coupling at all. The relaxed-stretch mechanism parallels the load-redistribution explanation of steered postbuckling stiffness [23, 39].

6.3 Limitations and scope

- **Buildability.** The designs are curvature-feasible idealized tow fields. The curvature limit holds pointwise (radius $\geq 19.9\,\mathrm{mm}$ on an independent grid), but the field spreads tows by more than a decade (Sec. 5.7). Laying it requires course-level tow cut/add, and the as-built model shifts the frequency by 0.96–1.12 $\omega_{1,\mathrm{UD}}$ and the mass by 0.97–1.12 of UD. A buildable design at exactly held frequency would include a thickness/defect field tied to $\theta(\boldsymbol{x})$ in the optimization. The defect model here is a continuum estimate; ply-level gap/overlap geometry [36, 34], tow width and coverage strategy, and process effects (wrinkling, resin-rich zones) are not resolved.
- **Laminate.** The plate designs are one steered layer, with $\boldsymbol{B} = \boldsymbol{0}$ but unbalanced ($A_{16} \neq 0$). A multi-ply laminate with balanced, independently steered layers is not studied. The beam design with no tow limit turns to about 83° in narrow zones of the 20 mm-wide strip.
- **Amplitude range.** The design quantity is the small-amplitude cubic coefficient. The cubic SSM reproduces the full-order EPMC detuning to within 5% over the amplitudes EPMC reached ($a/h \leq 0.21$ for UD, ≤ 0.35 for the steered design), and the jump-onset amplitudes reported at $\zeta = 2\%$ lie inside these ranges (those at $\zeta = 4\%$ in Sec. 5.7 do not). Higher amplitudes need the quintic SSM or EPMC (Appendix D).
- **Single mode, no internal resonance.** On the full multi-mode plate the steered design's lowest frequencies are in ratios 1:2.49:2.86:4.16 (`mode_ratios.py`). The nearest commensurability (3:1 with the third mode) is detuned by 5%, and that mode is antisymmetric about the center. The six-mode modal time-integration model contains these modes and would show an internal-resonance exchange; its sweeps follow the single-mode forced response instead (Fig. 6b,c).
- **Quasi-static in-plane field.** The relaxed cubic slaves the in-plane field statically. Proposition 2 bounds the error by $4\omega_1^2/\omega_{\mathrm{in}}^2 = 3.2 \times 10^{-3}$, and the full SSM, which retains in-plane inertia, agrees to 1.2%.
- **Optimality.** The optimizer is local (multi-start SLSQP); two of six starts stall (Table 2). The pointwise ceiling of Sec. 5.4 is itself a local optimum, so it estimates, but does not bound, what a single-ply field can do on this plate.
- **Scope of the robustness study.** Five plates, one material, one thickness ($a/h = 100$, thin von Kármán), and the fundamental mode only. Thick plates, shells, other modes and damping that is itself designed are not studied.

6.4 Outlook

The same reduction and adjoint carry over directly to three extensions:

- *prescribing* the backbone (a target frequency–amplitude law) instead of minimizing the hardening;
- designing tow path and thickness/defect field together, so that a buildable laminate holds the frequency exactly;
- steering stability boundaries, such as supersonic panel flutter, whose nonsymmetric aeroelastic operator enters the same first-order pencil; the flutter and post-flutter response of plates with prescribed curvilinear fibers has been analyzed [41, 42, 43], but not designed.

The mechanism of Sec. 3 predicts where each extension should help: wherever two functionals of the same stiffness are weighted by different strain fields, in proportion to the conflict between those fields.

7 Conclusions

The frequency and the geometric hardening of a laminated panel integrate the same rotated stiffness against two strain fields of the fundamental, the curvature and the relaxed stretch. Each density reads the fiber angle only through the anisotropic invariants and the local principal directions. Where the two fields disagree, a spatial fiber field can serve both and a uniform layup cannot. On a clamped beam this means compliant fibers at the inflection points. On a simply supported plate, at the unidirectional frequency and mass, it reduces the hardening by about two thirds in every model we checked. No uniform laminate is available at that frequency, and above it steering undercuts every uniform class surveyed. The lever vanishes for isotropic material, grows with anisotropy and with the conflict between the fields, and saturates once tows can turn on the fields' length scale. Because the jump-free amplitude scales as $J^{-1/2}$, the gain appears directly as a $1.66\times$ wider hysteresis-free range, independent of the damping level. Turning the idealized tow fields into buildable laminates requires tow cut/add, which our defect model suggests moves the design along, not off, the steered front.

Data and code availability. The code, result files and the scripts that generate every number, table and figure (Appendix E) are in the repository `https://github.com/jtempelman95/NLTopt` (tag `R1`; see `docs/paper/README_REPRODUCE.md`). The SSM, EPMC and time-integration scripts build on the author's harmonic-balance package PyHB, which is not public and is available on request. The design evaluator and the regeneration of every number from the committed result files do not need it.

Competing interests. The author declares no competing interests.

A Computation of the SSM coefficients

This appendix gives the exact recursion that produces every coefficient $\boldsymbol{w_m}$ (the geometry) and γ_{2k+1} (the reduced dynamics) used in the paper, matching the implementation line for line.

A.1 Master spectral data

Cast (9) in first order, $\boldsymbol{\mathcal{B}}\dot{\boldsymbol{z}} = \boldsymbol{\mathcal{A}}\boldsymbol{z} + \boldsymbol{F}(\boldsymbol{z})$, with $\boldsymbol{\mathcal{A}}, \boldsymbol{\mathcal{B}}$ as in Sec. 2.3. Solve the linear generalized eigenproblem $\boldsymbol{K}\boldsymbol{\varphi} = \omega_0^2 \boldsymbol{M}\boldsymbol{\varphi}$, mass-normalize $\boldsymbol{\varphi}^\top \boldsymbol{M}\boldsymbol{\varphi} = 1$, and set $\zeta_0 = \boldsymbol{\varphi}^\top \boldsymbol{C}\boldsymbol{\varphi}/(2\omega_0)$. The master pencil eigenpair solves $\boldsymbol{\mathcal{A}}\boldsymbol{v}_1 = \lambda_1 \boldsymbol{\mathcal{B}}\boldsymbol{v}_1$ with $\lambda_1 = -\zeta_0\omega_0 + i\omega_0\sqrt{1-\zeta_0^2}$; scale $\boldsymbol{v}_1$ so its displacement

block is $\boldsymbol{\varphi}$. For symmetric $\boldsymbol{M}, \boldsymbol{C}, \boldsymbol{K}$ the pencil is complex-symmetric, so the left eigenvector is $\boldsymbol{u}_1 = \bar{\boldsymbol{v}}_1$ (normalized $\boldsymbol{u}_1^* \boldsymbol{\mathcal{B}} \boldsymbol{v}_1 = 1$); $(\lambda_1, \boldsymbol{v}_1)$ is refined from the seed $([\boldsymbol{\varphi}; \lambda_1 \boldsymbol{\varphi}], \lambda_1)$ by Rayleigh-quotient inverse iteration, $\boldsymbol{y} = (\boldsymbol{\mathcal{A}} - \lambda_1 \boldsymbol{\mathcal{B}})^{-1} \boldsymbol{\mathcal{B}} \boldsymbol{v}_1$, $\lambda_1 \leftarrow \boldsymbol{v}_1^\top \boldsymbol{\mathcal{A}} \boldsymbol{v}_1 / \boldsymbol{v}_1^\top \boldsymbol{\mathcal{B}} \boldsymbol{v}_1$, which converges cubically. (A dense QZ factorization of the first-order pencil is used only for non-symmetric systems; for the laminated beam its master eigenvector lost up to ten digits at some ply angles.) Set $\boldsymbol{v}_2 = \bar{\boldsymbol{v}}_1$, $\lambda_2 = \bar{\lambda}_1$. Seed the coefficient tables at first order: $\boldsymbol{w}_{(1,0)} = \boldsymbol{v}_1$, $\boldsymbol{w}_{(0,1)} = \boldsymbol{v}_2$, $R^1_{(1,0)} = \lambda_1$, $R^2_{(0,1)} = \lambda_2$.

A.2 Right-hand side at each order

For a multi-index $\boldsymbol{m} = (m_1, m_2)$, $|\boldsymbol{m}| \geq 2$, the homological equation (10) has right-hand side $\boldsymbol{G}_{\boldsymbol{m}} = \boldsymbol{G}^{\mathrm{nl}}_{\boldsymbol{m}} - \boldsymbol{G}^{\mathrm{cr}}_{\boldsymbol{m}}$, both assembled from *lower* orders. The nonlinear part is the order-$\boldsymbol{m}$ term of $-\boldsymbol{F}(\boldsymbol{W}(\boldsymbol{p}))$, a multi-index convolution of the displacement blocks $\boldsymbol{w}^x$ over all non-trivial partitions,

$$\boldsymbol{G}^{\mathrm{nl}}_{\boldsymbol{m}} = -\sum_{\substack{\boldsymbol{a}+\boldsymbol{b}=\boldsymbol{m} \\ \boldsymbol{a},\boldsymbol{b}\neq\boldsymbol{0}}} \begin{bmatrix} \boldsymbol{B}_2(\boldsymbol{w}^x_{\boldsymbol{a}}, \boldsymbol{w}^x_{\boldsymbol{b}}) \\ \boldsymbol{0} \end{bmatrix} - \sum_{\substack{\boldsymbol{a}+\boldsymbol{b}+\boldsymbol{c}=\boldsymbol{m} \\ \boldsymbol{a},\boldsymbol{b},\boldsymbol{c}\neq\boldsymbol{0}}} \begin{bmatrix} \boldsymbol{T}_3(\boldsymbol{w}^x_{\boldsymbol{a}}, \boldsymbol{w}^x_{\boldsymbol{b}}, \boldsymbol{w}^x_{\boldsymbol{c}}) \\ \boldsymbol{0} \end{bmatrix}, \tag{35}$$

and the cross-term part is what $\mathrm{D}\boldsymbol{W}\,\boldsymbol{R}$ already contributes from known nonlinear reduced coefficients, $\boldsymbol{G}^{\mathrm{cr}}_{\boldsymbol{m}} = \boldsymbol{\mathcal{B}} \sum_{j=1,2} \sum_{\boldsymbol{a}+\boldsymbol{e}_j=\boldsymbol{m},\,|\boldsymbol{b}|\geq 2} a_j\, \boldsymbol{w}_{\boldsymbol{a}} R^j_{\boldsymbol{b}}$, evaluated over the already-computed $\boldsymbol{w}, R$.

A.3 Solving each order

Sweep $q = 2, \ldots,$ order, and within each q the indices $\boldsymbol{m} = (m_1, m_2)$ with $m_1 \geq m_2$ (the conjugate partner is filled by symmetry). Put $\Lambda_{\boldsymbol{m}} = m_1\lambda_1 + m_2\lambda_2$. Two cases:

- **Near-resonant** ($m_1 - m_2 = 1$, i.e. $\boldsymbol{m} = (k{+}1, k)$, where $\Lambda_{\boldsymbol{m}} = \lambda_1$ conservatively): the operator $\Lambda_{\boldsymbol{m}}\boldsymbol{\mathcal{B}} - \boldsymbol{\mathcal{A}}$ is (near-)singular along $\boldsymbol{v}_1$. Project onto the left eigenvector to read the reduced coefficient, then solve the deflated system for the geometry under the normal-form gauge $\boldsymbol{u}_1^* \boldsymbol{\mathcal{B}} \boldsymbol{w}_{\boldsymbol{m}} = 0$:

$$\boxed{\gamma_{2k+1} = R^1_{\boldsymbol{m}} = \boldsymbol{u}_1^* \boldsymbol{G}_{\boldsymbol{m}}}, \qquad (\Lambda_{\boldsymbol{m}}\boldsymbol{\mathcal{B}} - \boldsymbol{\mathcal{A}})\, \boldsymbol{w}_{\boldsymbol{m}} = \boldsymbol{G}_{\boldsymbol{m}} - \boldsymbol{\mathcal{B}}\boldsymbol{v}_1\, R^1_{\boldsymbol{m}}, \tag{36}$$

  followed by $\boldsymbol{w}_{\boldsymbol{m}} \leftarrow \boldsymbol{w}_{\boldsymbol{m}} - (\boldsymbol{u}_1^* \boldsymbol{\mathcal{B}} \boldsymbol{w}_{\boldsymbol{m}}) \boldsymbol{v}_1$. Under light damping $\Lambda_{\boldsymbol{m}} \neq \lambda_1$ exactly and the solve is nonsingular; at exact resonance one uses the least-squares pseudo-inverse and the same deflation.

- **Non-resonant**: set $R_{\boldsymbol{m}} = \boldsymbol{0}$ and invert directly, $\boldsymbol{w}_{\boldsymbol{m}} = (\Lambda_{\boldsymbol{m}}\boldsymbol{\mathcal{B}} - \boldsymbol{\mathcal{A}})^{-1} \boldsymbol{G}_{\boldsymbol{m}}$.

The reduced coefficients are $\{\gamma_{2k+1}\} = \{R^1_{(k+1,k)}\}$ (γ_3 at $k{=}1$, γ_5 at $k{=}2$, …), and the backbone is (11).

A.4 Worked coefficient: the cubic term

At order 3 the only near-resonant index is $\boldsymbol{m} = (2, 1)$. By (35) its forcing is the cubic force on the master eigenvectors plus the quadratic acting on the order-2 map, $\boldsymbol{G}_{(2,1)} = -\big[\, 3\,\boldsymbol{T}_3(\boldsymbol{v}_1, \boldsymbol{v}_1, \bar{\boldsymbol{v}}_1) + 2\boldsymbol{B}_2(\boldsymbol{v}_1, \boldsymbol{w}_{(1,1)}) + 2\boldsymbol{B}_2(\bar{\boldsymbol{v}}_1, \boldsymbol{w}_{(2,0)}) \,\big]$ (displacement block), and (36) gives $\gamma_3 = \boldsymbol{u}_1^* \boldsymbol{G}_{(2,1)}$. For a single-DOF Duffing $\ddot{x} + \omega_0^2 x + \varepsilon x^3 = 0$ the quadratic vanishes, the order-2 map is zero, and γ_3 reduces to the classical $\Omega = \omega_0\big(1 + \frac{3}{8}(\varepsilon/\omega_0^2)A^2\big)$. For the von Kármán plate the membrane coupling $\boldsymbol{B}_2$ (the $u \leftrightarrow w^2$ term) is nonzero, its order-2 slaving $\boldsymbol{w}_{(1,1)}, \boldsymbol{w}_{(2,0)}$ enters through $\boldsymbol{B}_2$ above, and $\operatorname{Im}\gamma_3$ equals $\frac{3}{2\omega_1}$ times the relaxed cubic γ of Eq. (13), the direct membrane cubic *minus* the in-plane relaxation, up to the $O(4\omega_1^2/\omega_{\mathrm{in}}^2)$ correction of Appendix B. The reduced coefficient (36), $\gamma_{2k+1} = \boldsymbol{u}_1^* \boldsymbol{G}_{(k+1,k)}$, is an adjoint projection, which is why its design sensitivity is inexpensive.

B The relaxed cubic is the SSM cubic (proof of Proposition 2)

Take (9) undamped with $\boldsymbol{B} = \boldsymbol{0}$ and split $\boldsymbol{x} = [\boldsymbol{q}; \boldsymbol{s}]$. The linear operators are block diagonal, so the master right eigenvector is $\boldsymbol{v}_1 = [\boldsymbol{\varphi}; \boldsymbol{0}; i\omega_1\boldsymbol{\varphi}; \boldsymbol{0}]$. Normalizing $\boldsymbol{u}_1^*\boldsymbol{\mathcal{B}}\boldsymbol{v}_1 = 1$ with $\boldsymbol{u}_1 = \bar{\boldsymbol{v}}_1$ gives the scaling $1/(2i\omega_1)$. The quadratic force has two parts. Its $\boldsymbol{s}$-rows are a symmetric form in $\boldsymbol{q}$, written $\boldsymbol{b}(\boldsymbol{q}, \boldsymbol{q})$ (the in-plane drive of the stretch). Its $\boldsymbol{q}$-rows are bilinear in $(\boldsymbol{q}, \boldsymbol{s})$. These are the two gradients of the single term $\int \boldsymbol{\varepsilon}_L^\top \boldsymbol{A}\boldsymbol{\varepsilon}_{NL}$. The cubic force acts in the $\boldsymbol{q}$-rows only.

Order 2. The forcing at $\boldsymbol{m} \in \{(2,0), (1,1), (0,2)\}$ has only $\boldsymbol{s}$-rows. Their coefficients are $1, 2, 1$ times $\boldsymbol{b}(\boldsymbol{\varphi}, \boldsymbol{\varphi})$, from expanding $(p_1 + p_2)^2$. The displacement-level homological equations are

$$(\boldsymbol{K}_{\text{in}} + \Lambda_{\boldsymbol{m}}^2 \boldsymbol{M}_{\text{in}})\, \boldsymbol{s}_{\boldsymbol{m}} = -c_{\boldsymbol{m}}\, \boldsymbol{b}(\boldsymbol{\varphi}, \boldsymbol{\varphi}), \qquad \Lambda_{(1,1)} = 0, \quad \Lambda_{(2,0)} = 2i\omega_1,$$

so $\boldsymbol{s}_{(1,1)} = -2\boldsymbol{K}_{\text{in}}^{-1}\boldsymbol{b}$ is *exactly* static, while $\boldsymbol{s}_{(2,0)} = -(\boldsymbol{K}_{\text{in}} - 4\omega_1^2\boldsymbol{M}_{\text{in}})^{-1}\boldsymbol{b}$.

Order 3. The resonant coefficient collects the cubic term ($3\times$) and the quadratic $\boldsymbol{q}$-rows acting on $\boldsymbol{s}_{(1,1)}$ and $\boldsymbol{s}_{(2,0)}$. With static $\boldsymbol{s}$, the combination is exactly $3(k_{nl} - 2\boldsymbol{g}^\top \boldsymbol{K}_{\text{in}}^{-1}\boldsymbol{g})$, in which $\boldsymbol{s}_{(1,1)}$ supplies two thirds of the relaxation and $\boldsymbol{s}_{(2,0)}$ one third. Projecting with $\boldsymbol{u}_1$ gives $\operatorname{Im}\gamma_3 = \frac{3}{2\omega_1}\gamma$.

Error. Expanding the resolvent, $(\boldsymbol{K}_{\text{in}} - 4\omega_1^2\boldsymbol{M}_{\text{in}})^{-1} = \boldsymbol{K}_{\text{in}}^{-1} + 4\omega_1^2\boldsymbol{K}_{\text{in}}^{-1}\boldsymbol{M}_{\text{in}}\boldsymbol{K}_{\text{in}}^{-1} + \dots$. This perturbs the $(2,0)$ third of the relaxation by at most the relative amount $\rho/(1-\rho)$, with $\rho = 4\omega_1^2/\omega_{\text{in}}^2$. Hence

$$\operatorname{Im}\gamma_3 = \frac{3}{2\omega_1}(\gamma + \epsilon), \qquad |\epsilon| \le \tfrac{1}{3}\left(2\boldsymbol{g}^\top \boldsymbol{K}_{\text{in}}^{-1}\boldsymbol{g}\right)\frac{\rho}{1-\rho}.$$

For the plate, $\rho = 3.2 \times 10^{-3}$, so $|\epsilon|/\gamma < 0.1\%$. In-plane inertia is negligible for the backbone. The residual differences between (13) and the full SSM in Sec. 5.5 come from the transverse basis, not from the condensation.

C Finite-element validation of the assumed-modes discretization

Why this is needed. The reductions above are verified against the analytic Duffing backbone and against full-order EPMC and direct time integration. Those confirmations, however, are of the *same* assumed-modes system solved at higher fidelity: they validate the SSM *reduction*, not the assumed-modes *discretization*. A steered laminate is spatially heterogeneous and, off the principal axes, develops bend–twist couplings D_{16}, D_{26}; whether a handful of global sine modes resolve its response is a separate question. We settle it with an independent finite-element (FE) model and the identical backbone coefficient.

C.1 FE model

We take a flat, symmetric ($\boldsymbol{B} = \boldsymbol{0}$) plate on a uniform $n_x \times n_y$ rectangular mesh, bending and membrane discretized independently (they decouple linearly for $\boldsymbol{B} = \boldsymbol{0}$).

Bending (Kirchhoff). The non-conforming 12-DOF ACM rectangular element carries nodal degrees of freedom $\boldsymbol{d}_e = \{w,\ w_{,y},\ -w_{,x}\}$ at the four corners. On the element (local $s, t \in [0,1]$) the deflection is the 12-term polynomial

$$w(s,t) = \boldsymbol{p}^\top \boldsymbol{a}, \quad \boldsymbol{p} = [1, s, t, s^2, st, t^2, s^3, s^2t, st^2, t^3, s^3t, st^3]^\top; \tag{37}$$

enforcing the nodal DOF gives $\boldsymbol{d}_e = \boldsymbol{C}\boldsymbol{a} \Rightarrow \boldsymbol{a} = \boldsymbol{C}^{-1}\boldsymbol{d}_e$. With the curvature $\boldsymbol{\kappa} = [w_{,xx}, w_{,yy}, 2w_{,xy}]^\top = \boldsymbol{B}_b\boldsymbol{C}^{-1}\boldsymbol{d}_e$ ($\boldsymbol{B}_b$ the side-scaled second derivatives of $\boldsymbol{p}$), and 4×4 Gauss quadrature,

$$\boldsymbol{K}_e^b = \int_{\Omega_e} \boldsymbol{C}^{-\top}\boldsymbol{B}_b^\top\, \boldsymbol{D}(\theta_e)\, \boldsymbol{B}_b\boldsymbol{C}^{-1}\, \mathrm{d}\Omega, \quad \boldsymbol{M}_e^b = \rho h \int_{\Omega_e} \boldsymbol{C}^{-\top}\boldsymbol{p}\boldsymbol{p}^\top\boldsymbol{C}^{-1}\, \mathrm{d}\Omega, \tag{38}$$

field	model	f_1 [Hz]	γ	detuning @ a/h=0.4
UD [0] (homogeneous)	FE (28×21)	256.0	2.878×10^{10}	11.06%
	assumed-modes	256.3	2.842×10^{10}	10.96% (1%)
steered (strong)	FE (28×21)	334.4	2.515×10^{10}	5.73%
	assumed, 8 modes	358.9	2.546×10^{10}	5.30% (7%)
	assumed, 24 modes	345.7	2.584×10^{10}	5.65% (2%)

Table 6: Assumed-modes vs. FE backbone (parenthesized: detuning error vs. FE). "N modes" is the number of transverse modes in total (the N lowest), with 3×3 in-plane modes; the design evaluator uses 64 transverse ($m, n \le 8$) and 6×6 in-plane modes (Table 1). This test therefore checks the transverse discretization and the nonlinear FE model on a smooth steered field (20°–65°); the in-plane resolution of the actual design is checked separately by the FE evaluation of the optimized field (ratio 0.364, Sec. 5.5). Homogeneous: the two agree to 1%, validating the FE nonlinear model. Strong steering: the frequency is a slowly-converging Ritz upper bound (+7% at 8 modes, +3% at 24), but the *normalized detuning* — the design quantity — matches FE to 7% at 8 modes and 2% at 24.

$\boldsymbol{D}(\theta_e) = \frac{h^3}{12}\bar{\boldsymbol{Q}}(\theta_e)$ evaluated per element, so D_{16}, D_{26} are retained exactly.

Membrane (Q4). The in-plane fields use the 8-DOF bilinear quadrilateral; with the membrane strain–displacement $\boldsymbol{B}_m$ and $\boldsymbol{A}(\theta_e) = h\,\bar{\boldsymbol{Q}}(\theta_e)$, $\boldsymbol{K}_e^{\text{in}} = \int_{\Omega_e} \boldsymbol{B}_m^\top \boldsymbol{A}(\theta_e)\boldsymbol{B}_m \,\mathrm{d}\Omega$.

Boundary conditions. Navier simple support: $w = 0$ on the boundary (bending) and $u = v = 0$ on the boundary (membrane) — matching the sine in-plane modes, which vanish on the edges. *Self-validation:* for a specially-orthotropic UD [0] plate the FE reproduces the closed form $\omega_{mn} = \pi^2\sqrt{[D_{11}(m/a)^4 + 2(D_{12}+2D_{66})(m/a)^2(n/b)^2 + D_{22}(n/b)^4]/\rho h}$ to $f_1 = 256.0$ vs. 256.3 Hz, converging monotonically with mesh (`examples/plate_fe_check.py`).

C.2 Backbone by static condensation on the mesh

We compute the *same* backbone coefficient the plate results use — Eq. (13) — but FE-discretized. Solve $\boldsymbol{K}^b\boldsymbol{\varphi} = \omega_1^2\boldsymbol{M}^b\boldsymbol{\varphi}$ for the fundamental mode (unit peak deflection); its von Kármán membrane strain is $\boldsymbol{\varepsilon}_{NL} = [\frac{1}{2}w_{,x}^2, \frac{1}{2}w_{,y}^2, w_{,x}w_{,y}]^\top$, with the gradients read off the ACM element. Assembling $\boldsymbol{g} = \int \boldsymbol{B}_m^\top \boldsymbol{A}\,\boldsymbol{\varepsilon}_{NL}$, $\mathrm{knl} = 2\int \boldsymbol{\varepsilon}_{NL}^\top \boldsymbol{A}\,\boldsymbol{\varepsilon}_{NL}$ and $K_b = \boldsymbol{\varphi}^\top \boldsymbol{K}^b\boldsymbol{\varphi}$, static condensation of the fast in-plane fields gives

$$\gamma = \mathrm{knl} - 2\,\boldsymbol{g}^\top(\boldsymbol{K}^{\text{in}})^{-1}\boldsymbol{g}, \qquad \Omega(a)/\omega_1 = 1 + \beta\,(a/h)^2, \quad \beta = \tfrac{3}{8}\,(\gamma/K_b)\,h^2. \tag{39}$$

This costs *one* sparse solve — no continuation or time integration — because the condensation *is* the order-2 SSM slaving (App. A) evaluated on the mesh. The identical formula on the sine basis returns the assumed-modes backbone, so any difference is purely discretization (`examples/plate_fe_backbone.py`).

C.3 Results

Table 6 and Fig. 8 compare the two discretizations for a homogeneous and a strongly steered field ($\theta = 20° + 45°\sin\frac{\pi x}{a}\sin\frac{\pi y}{b}$, sweeping 20°–65°).

Two conclusions. *(i)* For a homogeneous laminate the FE and assumed-modes backbones agree to 1%, validating the FE nonlinear model (membrane, coupling, condensation). *(ii)* For strong steering the assumed-modes frequency is a Ritz upper bound that converges slowly, but the *normalized backbone detuning* $\beta \propto \gamma/K_b$ is far more robust — a ratio of two integrals of the *same*

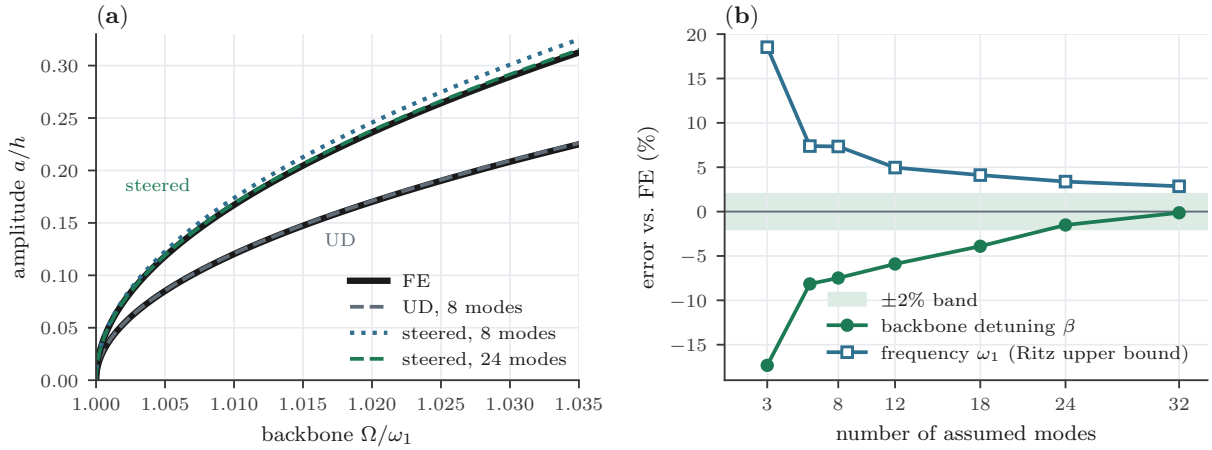


Figure 8: **The assumed-modes backbone agrees with an independent finite-element model. (a)** Amplitude vs. backbone frequency (same orientation as Figs. 2 and 6); black: FE. For the homogeneous UD plate the 8-mode assumed-modes backbone (gray dashed) coincides with FE, validating the FE nonlinear model; for the strongly steered plate the assumed-modes backbone converges to FE as modes are added (8 modes dotted, 24 modes green dashed, overlying FE). **(b)** Ritz convergence toward FE (steered): the normalized backbone detuning enters the ±2% band by $\sim$24 modes, whereas the absolute frequency is a slowly-converging Ritz upper bound. The design quantity (relative detuning) is the well-resolved one.

slightly-inaccurate mode, so the mode-shape error largely cancels, leaving a 2% discrepancy at 24 modes. The relative detuning results the designs are built on are therefore FE-accurate to a few percent; absolute frequencies of strongly steered designs should be read from FE or an enriched ($\gtrsim$ 24-mode) expansion. The paper's design evaluator uses 64 transverse modes, and the optimized design itself is evaluated on this FE model in Sec. 5.5 (detuning ratio 0.364 vs. 0.359 from the full SSM). A displacement-based von Kármán FE would replace the assumed modes throughout at higher cost, reduced by the same SSM.

D Calibration against prior work, and SSM- vs EPMC-driven optimization

We reproduce two published results of [15] as a check of the reduction and its sensitivities, and compare SSM- and EPMC-driven optimization on one benchmark. *(i)* The backbone sensitivity of two coupled oscillators predicts the perturbed backbone to within about 3×10^{-5} in Ω over its soften-then-harden shape (the frequency first falls, then turns back up near an RMS amplitude of 1). Capturing the turn-back needs a high-order SSM; we use order seven, where [15] use order five with an analytic adjoint (Fig. 9). The damped eigenvalue $\lambda = -0.0191 \pm 0.6177i$ matches exactly. *(ii)* A clamped von Kármán beam is shaped by an initial curvature to a prescribed softening backbone, and full-order EPMC confirms the optimized SSM backbone to $< 0.1\%$ (Fig. 10). Two departures are noted. First, we use the axially condensed shallow-arch model rather than a three-DOF-per-node element. Second, we place the target amplitudes inside the cubic SSM's verified window. Eq. (33) of [15], taken literally, places a same-sign quadratic coupling on both masses, which is not energy-consistent; the potential-derived spring we use leaves the reported eigenvalue unchanged.

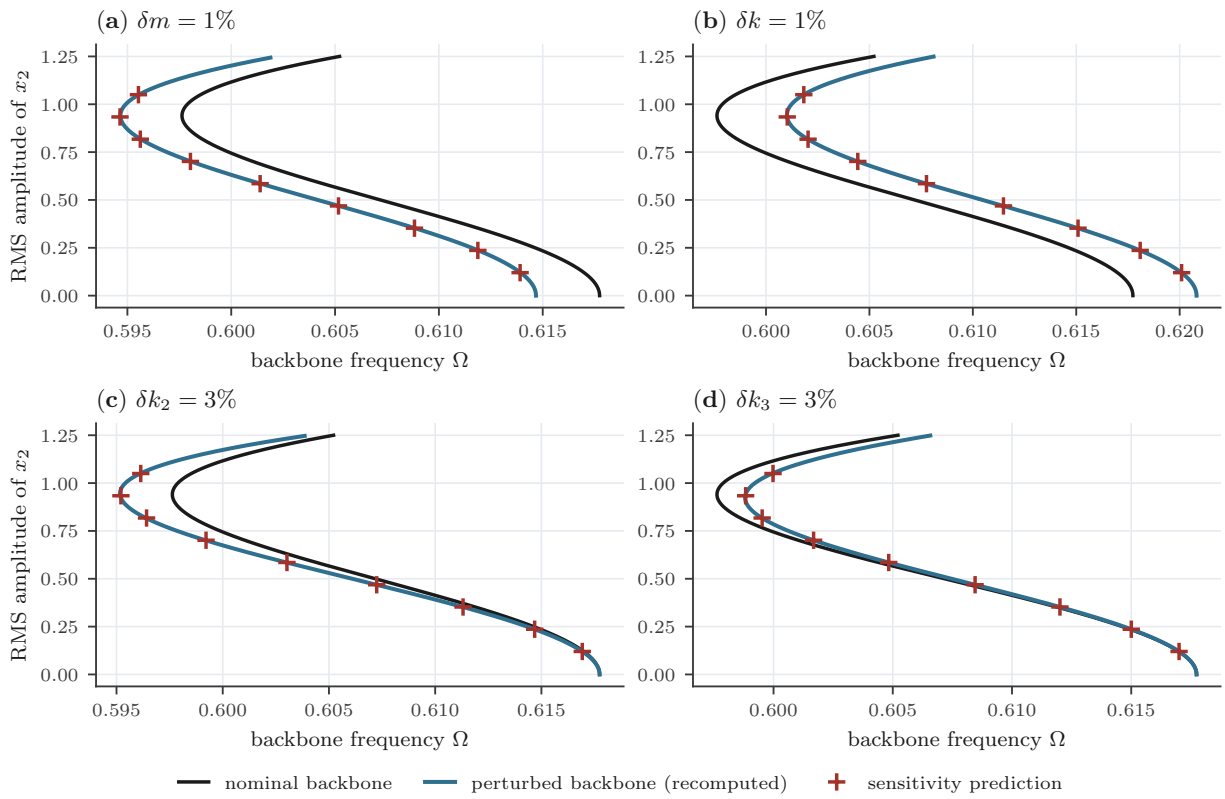


Figure 9: **The backbone sensitivities reproduce a published benchmark.** Calibration against [15], Fig. 2: backbone-sensitivity validation of two coupled oscillators. Each panel perturbs one parameter (m, k: 1%; k_2, k_3: 3%). Black: nominal backbone; blue: perturbed backbone, recomputed; red crosses: first-order sensitivity prediction from the nominal backbone. The predictions land on the recomputed backbones through the soften-then-harden turn-back, which the seventh-order SSM captures.

Validity window of the cubic SSM. Over the amplitudes reached by the full-order EPMC continuation ($a/h \leq 0.21$ for UD, ≤ 0.35 for the steered design; Fig. 6a), the cubic SSM backbone reproduces the EPMC detuning $\Omega/\omega_1 - 1$ to within 5%. The jump thresholds compared in Sec. 5.6 ($a_c/h = 0.208$ and 0.347) lie inside these ranges; we make no claim beyond them, where higher-order terms bend the true backbone.

EPMC as a design engine. The extended periodic motion concept [9] augments the equation of motion with a self-excitation that offsets damping, $\boldsymbol{M}\ddot{\boldsymbol{q}} + (\boldsymbol{C} - 2\delta\Omega\boldsymbol{M})\dot{\boldsymbol{q}} + \boldsymbol{K}\boldsymbol{q} + \boldsymbol{f}_{nl} = \boldsymbol{0}$, and continues the harmonic-balance residual $\boldsymbol{R}(\boldsymbol{z}; a, \theta) = \boldsymbol{0}$ in amplitude. Its design sensitivity at a converged point follows from the implicit-function theorem with the Newton Jacobian already assembled, through one adjoint solve per amplitude, $\mathrm{d}\Omega/\mathrm{d}\theta = -\boldsymbol{w}^{\top}\partial_{\theta}\boldsymbol{R}$ with $\boldsymbol{J}_z^{\top}\boldsymbol{w} = \boldsymbol{e}_{\Omega}$. $\partial_{\theta}\boldsymbol{R}$ reuses the same $\partial\bar{\boldsymbol{Q}}/\partial\theta$ as the SSM gradient. The two engines are complementary. The SSM gives a smooth, analytic objective at a cost independent of amplitude and harmonic count, and is the search engine here. EPMC is exact at any amplitude it reaches, captures folds and internal resonances, and serves as the validation of record. On the benchmark below, both find nearly the same optimum, and the SSM backbone is about two orders of magnitude cheaper per evaluation.

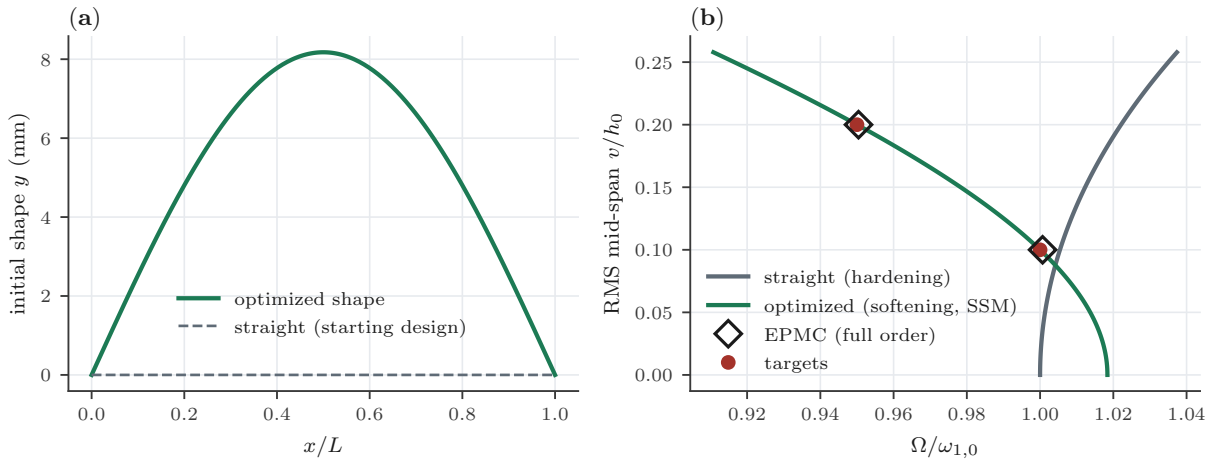
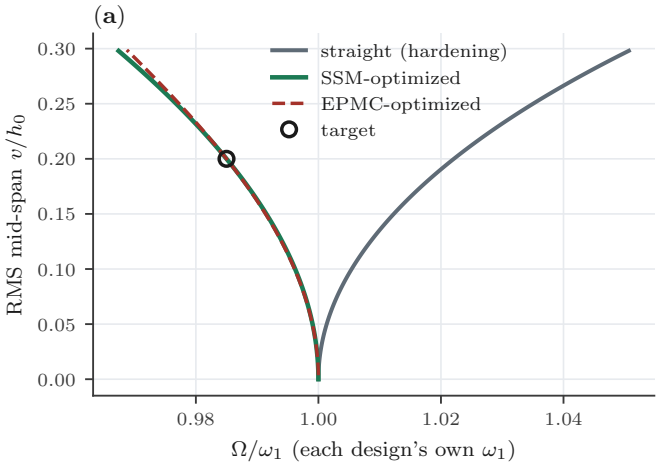


Figure 10: **The SSM design loop reproduces a published backbone-shaping result.** Calibration against [15], Fig. 3: a von Kármán beam shaped to a prescribed softening backbone. This problem shapes the beam with up to two harmonics (the optimum uses only the first) to pass through two prescribed backbone points; it is a different problem from that of Fig. 11, which tunes one amplitude to one target. **(a)** Optimized initial shape $y(x)$ and the straight starting design. **(b)** Backbones, normalized by the first frequency $\omega_{1,0}$ of the initial straight beam: the straight (hardening) beam becomes a curved (softening) one threading both target points; full-order EPMC (diamonds) confirms the optimized SSM backbone to $< 0.1\%$.

D.1 SSM- and EPMC-driven optimization on the same benchmark

To compare the two engines as *optimizers*, we solve the same design problem with each on the published von Kármán-beam backbone-shaping benchmark of [15] (their Sec. 5.2 / Fig. 3): a clamped–clamped beam whose backbone is shaped from hardening to a prescribed *softening* by an initial curvature $y = A_1 \sin(\pi x/L)$. The straight beam hardens (+2.2% relative detuning at RMS $= 0.2h_0$); the curvature injects a quadratic nonlinearity whose slaved response drives the SSM cubic coefficient negative, softening the backbone (it also stiffens the beam linearly, so the curvature-invariant measure is the detuning relative to each design's *own* ω_1). We design A_1 to hit a -1.5% softening target, driving the optimizer once with the SSM backbone and once with the full-order EPMC backbone. Both reach the target exactly. The optima are $A_1 = 8.03\,\text{mm}$ (SSM) and $8.26\,\text{mm}$ (EPMC), a 2.8% difference that is the cubic SSM's truncation error at RMS $= 0.2h_0$. The SSM backbone costs about two orders of magnitude less per evaluation (Fig. 11). This is the intended division of labor: the SSM adjoint does the search cheaply, and full-order EPMC confirms the optimum.



(b)

	SSM	EPMC
optimal A_1 (mm)	8.03	8.26
detuning at $0.2h_0$	−1.50%	−1.50%
backbone evaluations	12	10
wall time (s)	0.059	11.2
cost per backbone	1×	≈ 230×

Figure 11: **SSM- and EPMC-driven optimization reach nearly the same optimum, at very different cost.** Optimizing the same design with both frameworks (benchmark of [15]). Here one curvature amplitude A_1 is tuned to one softening target (a different problem from Fig. 10). **(a)** The straight beam's backbone hardens (leans right); the optimized initial curvature softens it (leans left) through the softening target; the SSM- and EPMC-optimized backbones nearly coincide. **(b)** The two optima and their cost; the SSM optimum differs from the full-order one by the cubic SSM's truncation error at RMS $= 0.2h_0$, at about two orders of magnitude lower cost per backbone evaluation (wall times are indicative only).

E Verification and reproducibility

Every computed result quoted in the main text and in Appendices A–C is a macro generated from the result files by `examples/paper_numbers.py` (`docs/paper/numbers.tex`), or a table written by a listed script; the few rounded bounds and estimates stated in words (e.g. "less than 0.02", the bang-bang estimate 0.4–0.5) follow from those values. The calibration values of Appendix D (the replicated benchmarks of [15]) are printed by the scripts listed for them below. Table 7 maps each result to the script that produces it. Unit tests are in `tests/` (`pytest`).

Table 7: Result → script map.

result	script (in `examples/` unless a path is given)
relaxed evaluator, gradient, tests	`nltopt/relaxed_plate.py`, `tests/test_relaxed_plate.py`
beam test (Sec. 5.1, Fig. 2)	`beam_symmetric_steering.py`, `fig_beam.py`
B-coupled beam (comparison)	`steered_beam_amplitude_robust.py`
achievable sets (Sec. 5.2, Fig. 3)	`levers_study.py`, `fig_levers.py`
mechanism, front, dose (Figs. 4, 5)	`mechanism_study.py`, `fig_mechanism.py`, `fig_limits.py`
resolution and pointwise ceiling	`ceiling_study.py`
full-order confirmation (Sec. 5.5), jump (Sec. 5.6)	`confirm_heldfreq.py` (SSM, thresholds, sweeps), then `epmc_check.py` (merges the EPMC backbones into the same file), `confirm_fine_sweep.py`, `fig_confirm.py`
independent FE check of the design (Table 4)	`fe_check_design.py`
basis convergence (Table 1)	`conv_table.py`
continuous-stack competitor (Sec. 5.2)	`stack_continuous.py`
frequency multiplier, linear spectra (Secs. 5.3, 6)	`mechanism_lambda.py`, `mode_ratios.py`
all quoted numbers and generated tables	`paper_numbers.py`
FE validation (App. C)	`plate_fe_backbone.py`, `plate_fe_fig.py`
calibration (App. D)	`coupled_osc_sensitivity.py`, `beam_backbone_opt.py`, `ssm_vs_epmc_opt.py`
matched-frequency comparison (Sec. 5.5, Fig. 3b)	`r1_matched_frequency.py`
other plates, conflict (Sec. 5.7, Fig. 7a,b)	`r1_generality.py`, `r1_conflict.py`
damping level and law (Fig. 7c)	`r1_damping.py`
as-built tow defects (Sec. 5.7, Fig. 7d,e)	`r1_asbuilt.py`, `fig_robust.py`
gradient, independent-grid and multi-start checks (Table 2)	`r1_verification.py`

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